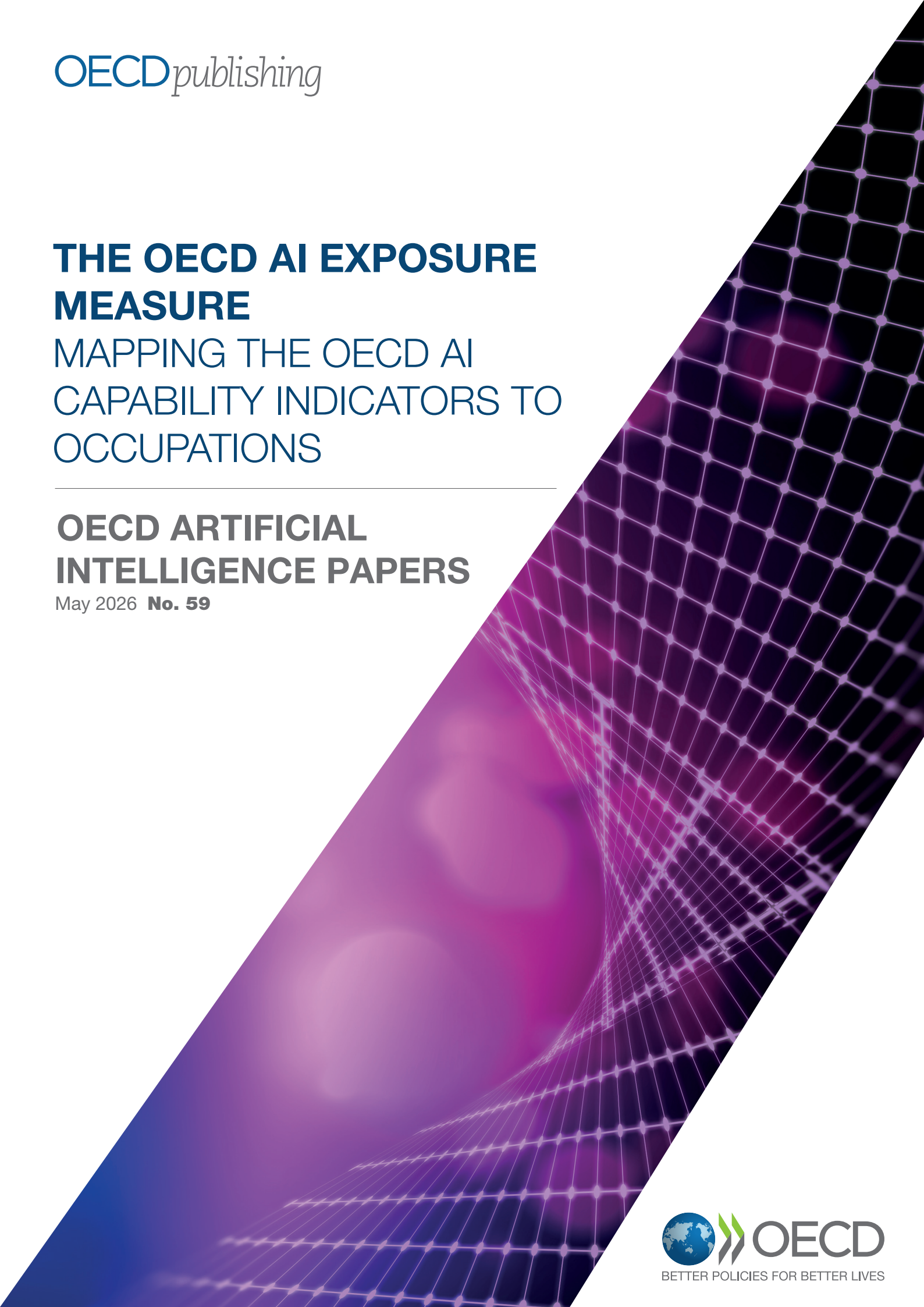
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THE OECD AI EXPOSURE MEASURE

MAPPING THE OECD AI CAPABILITY INDICATORS TO OCCUPATIONS

OECD ARTIFICIAL INTELLIGENCE PAPERS

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OECD Artificial Intelligence Papers

The OECD AI Exposure Measure

Mapping the OECD AI Capability Indicators to Occupations



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Revised version, July 2026. The following changes have been made:

The revision concerns only the vision capability domain, where the figures now reflect the correct level of current AI vision capability (3). Because the overall index sums the gaps across all nine capability domains, this also produces small changes to the affected vision figures and to the Overall AI Capability Gap Index wherever they appear. The individual changes are set out below. On p.6 in paragraph 2, the value of 24 was replaced by 23. On p.6 in the last paragraph continuing onto p.7, the values for some occupations changed slightly in the decimal place. The same changes were applied to paragraph 2 on p. 29. On p.7 in paragraph 3, the value for vision of 0.8 was replaced by 0.2. The same change was applied to the first paragraph on p. 30. On p.28, the last paragraph now clarifies that, while the theoretical upper bound remains 36, the effective maximum given current AI capability levels is 23. On p.29, Table 6.3 reflects the corrected values for vision and the Overall AI Capability Gap Index, which changed minimally. On p.31 in Table 6.4a, the vision numbers were reduced by 1. On p. 32 in Table 6.4b in the Vision section, the list of occupations changed, but the text remained the same. On p.33 in Table 6.5 some of the values changed slightly in the decimal place. On p.35 in Table 6.6, the values in the first column changed slightly with no resulting change to the text. On p.39, Table 7.1 reflects the corrected values for vision and the Overall AI Capability Gap Index, which changed minimally; the new numbers are reflected in the second paragraph on that page.

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This publication contributes to the OECD's Artificial Intelligence in Work, Innovation, Productivity and Skills (AI-WIPS) programme, which provides policy makers with new evidence and analysis to keep abreast of the fast-evolving changes in AI capabilities and diffusion, and their implications for the world of work. The programme aims to help ensure that adoption of AI in the world of work is effective, beneficial to all, people-centred and accepted by the population at large. AI-WIPS is supported by the German Federal Ministry of Labour and Social Affairs (BMAS) and will complement the work of the German AI Observatory in the ministry's Policy Lab Digital, Work & Society. For more information, visit https://oecd.ai/workinnovation-productivity-skills and https://denkfabrik-bmas.de/.

Executive summary

This paper develops a new OECD measure of occupational AI exposure based on the OECD AI Capability Indicators. The measure is designed to respond to a central policy need: understanding how AI is likely to affect the economy, the workforce and skill demand over the next 5 to 10 years in a way that is forward-looking, transparent, updateable and grounded in human abilities. It is forward-looking because it measures potential exposure relative to the current AI frontier before diffusion is complete. It is transparent because it separates the assessment of AI capabilities from the assessment of occupational capability demands and makes each side of the comparison explicit. It is updateable because the AI Capability Indicators and the occupational mappings can be revised separately as evidence changes and combined with explicit scenarios about future AI trajectories. And it is broadly linked to human abilities because it is built around a structured framework spanning nine cognitive, social and physical capability domains. This makes the resulting measure more policy-relevant for analysing the evolving implications of AI for work, skills and education.

The paper operationalises this approach by mapping the nine OECD AI Capability Indicators to occupational requirements in O*NET and constructing an occupational AI Capability Gap Index that measures the gap between the capabilities demanded by each occupation and current AI capabilities, first for each capability domain and then for the occupation as a whole. Since occupational capability demands can reach level 5 at their highest, while current AI systems perform only at capability levels 2 or 3 on the five-point scale, gap values for a single capability domain range from 0 to 2 or from 0 to 3, depending on the domain. A value of 0 means that AI has already reached the level required by the occupation in that domain, whereas values of 2 or 3 indicate that a substantial gap remains. The total index sums the gaps across all nine indicators, each weighted by its importance to the occupation, and, given current AI capability levels, has an effective maximum of 23. Total index values equal or close to 0 indicate that current AI systems already match, or are close to matching, the capability profile required for an occupation and therefore imply relatively high AI exposure and automation potential. Index values greater than 0 indicate that important capability gaps remain in some parts of the job, though not necessarily across all, which implies scope for human-AI collaboration and augmentation. Higher index values can also be understood as reflecting the breadth of the occupational role as currently structured rather than as a claim that individual incumbents are broadly beyond AI assistance or substitution. Moreover equal movements in the total index need not imply equivalent forms of technological change, since progress may occur unevenly across domains and the substantive implications of narrowing a social or perceptual gap may differ from narrowing a language or metacognitive one.

The index allows exposure to be interpreted in a more nuanced way than existing measures of automation risk. It is also designed to distinguish between different forms of AI exposure, including exposure to language- and reasoning-based systems on the one hand, and to robotics, machine vision and embodied AI on the other. Consistent with other AI exposure measures, the new index estimates potential exposure to AI capabilities rather than actual AI adoption or realised labour-market effects, which also depend on diffusion and notably the cost of AI and robotics systems (especially as compared to the cost of human operators), organisational update, regulation and social choice.

The results show substantial variation across occupations and occupational groups. At the level of major occupational groups, current AI capabilities are closest to occupations characterised by routine information-processing and administrative tasks. Office and administrative support occupations record the

lowest total AI Capability Gap index, at 0.8, followed by production occupations (1.8), food preparation and serving related occupations (2.4), and sales and related occupations (2.6). By contrast, the highest gap values are observed in occupations that rely more heavily on judgement, interpretation, responsibility, social interaction, and physical activities in non-standardised environments, including community and social service occupations (6.4), legal occupations (5.8), educational instruction occupations (5.8), healthcare practitioners and technical occupations (5.6), management occupations (5.5) and protection occupations (5.2), followed by science (4.8), arts, sports and media (4.1), and installation, maintenance and repair (3.5) occupations. Overall, these patterns suggest that AI exposure is currently highest in occupations with more standardised, codifiable and repetitive task content, including routine inspection and monitoring tasks, and lower in occupations where contextual judgement, interpersonal understanding, complex decision making and dexterity remain central. These results should not be read as direct predictions of replacement or immunity from AI-driven change, but as indicators of how close current AI capabilities are to the capability profiles required in different occupations.

The analysis also shows that AI exposure should not be treated as a single dimension. The extension of the mapping to the full occupational space highlights that different occupations are exposed to different combinations of AI capabilities. Language-based and reasoning-based AI are likely to matter most in occupations with high demands in language, social interaction, problem solving, metacognition and knowledge-intensive work. By contrast, embodied AI, robotics and machine vision are likely to matter more in occupations where vision, manipulation and robotic intelligence are central. The occupational space is therefore not divided simply into high- and low-exposure jobs. Rather, occupations differ according to which capabilities matter most: some are exposed primarily to generative and reasoning systems, others to physical automation and machine perception, and some to combinations of both. As a result of the multi-dimensional nature of this OECD measure, it overlaps most with narrower exposure measures for occupations primarily relying on routine language and information processing work, but diverges for occupations where social, perceptual or embodied capabilities are central.

Across capability domains, the results show that the largest average gaps between current AI capabilities and occupational requirements are in social interaction, problem solving and metacognition (all 1.1). These are followed by the physical domains of manipulation (0.7) and robotic intelligence (0.6), which may prove the most stubborn to close. By contrast, the smallest average gaps are observed in creativity (0.1), vision (0.2), language (0.4) and knowledge, learning and memory (0.5). (The creativity result should be read cautiously, since the upper end of the scale refers to exceptional creativity that is rarely required of occupations as a whole, which compresses the relevant range compared to the other capability scales.) These results suggest that current AI capabilities are closest to occupational requirements involving language processing, codified knowledge, visual perception and some forms of content generation, while remaining further from capabilities related to dexterity, contextual reasoning, social understanding, and complex judgement. Within domains, the largest gaps are concentrated in occupations that depend heavily on interpersonal judgement, context-specific interpretation, and physical activities in non-standardised environments. For example, occupations such as judges, chief executives, clinical and counselling psychologists, mental health counsellors, and marriage and family therapists rank among the least exposed in social interaction and problem solving. In manipulation, firefighters, surgical assistants, obstetricians and gynaecologists, and urologists remain among the least exposed, while firefighters also appear among the least exposed in robotic intelligence, alongside police and sheriff's patrol officers, athletes, and commercial divers.

Taken together, the findings show the value of a capability-based OECD AI Exposure Measure for workforce policy. By linking AI progress directly to occupational capability demands, the measure provides a transparent and updateable basis for understanding how AI may reshape work. It supports a more differentiated view of exposure than existing one-dimensional indicators, highlights where current AI systems are already approaching occupational requirements, and identifies where major human capabilities remain difficult to replicate. As such, it offers a useful foundation for future analysis of task-

level transformation, changing skill demand, and the implications of AI for education, training and labour market policy. It also provides a useful bridge to broader foresight and macroeconomic analysis. In particular, the measure could be used to explore the labour-market and economic implications of the OECD's AI scenarios through 2030 and serve as input into macroeconomic models of AI-driven productivity and structural change, with scope for future country-level applications.

1 Introduction

Artificial intelligence (AI) is advancing rapidly across a wide range of domains, raising important questions about how technological change may affect jobs, skills and labour markets. For policymakers, the key challenge is not only to understand which occupations are currently exposed to AI, but also to anticipate how that exposure may change as AI capabilities improve. Existing measures provide useful insights, but none of them combines a forward-looking, transparent, updateable and broadly skill-linked framework for understanding how AI may affect work.

This paper develops a new measure of occupational AI exposure using the OECD AI Capability Indicators. These indicators provide a structured framework for assessing AI capabilities across nine domains of human ability and allow them to be compared directly with the capability demands of occupations. This makes it possible to estimate the extent to which AI is catching up with the requirements of different jobs. The resulting measure, called the AI Capability Gap Index, captures potential exposure in terms of capability proximity, though it does not, on its own, measure actual AI adoption in workplaces or predict realised employment outcomes.

The AI Capability Gap Index is distinctive because it brings together four features that are not usually combined in existing measures. It is forward-looking, because it assesses potential exposure relative to the current AI frontier before diffusion is complete. It is transparent, because it separates the assessment of AI capabilities from the assessment of occupational capability demands. It is updateable, because both sides of the comparison can be revised as evidence changes and used in scenario analysis. And it is broadly linked to human abilities, because it compares AI progress to a structured framework of occupational capability demands rather than to a single technology proxy.

To operationalise the index, the paper maps the OECD AI Capability Indicators to occupational requirements using O*NET. A sample of occupations was assessed by OECD analysts and refined through a consensus process, before extending the mapping to all occupations with the support of large language models and further validation. The resulting measure provides a new basis for analysing patterns of occupational exposure and their implications for skills, workforce policy, and future labour-market analysis.

The results show that current AI exposure is highest in occupations with more standardised and codifiable task content, that important gaps remain in domains such as manipulation, social interaction, problem solving and metacognition, and that exposure is multidimensional rather than reducible to a single ranking of jobs. Compared with narrower exposure measures, the OECD index overlaps most in routine language and information-processing work, but diverges more sharply in occupations where social, perceptual or embodied capabilities are central.

The paper first reviews existing approaches to measuring AI exposure, then presents the OECD AI Capability Indicators and the mapping methodology, and finally discusses how the resulting measure can support analysis of AI's implications for labour market, skills and education policies.

2 Evaluating the impact of AI on jobs

What measures of AI exposure are out there

A long-standing strand of economic research on skill-biased technical change explains how technological change has shaped labour demand and wage inequality over time (Acemoglu, 2002[1]); (Goldin and Katz, 2008[2]). Related task-based work, including (Autor, Levy and Murnane, 2003[3]) (Autor, Katz and Kearney, 2006[4]) and (Acemoglu and Autor, 2011[5]), sharpened the idea that technological change affects jobs through the task content of work. (Autor and Thompson, 2025[6]) and (Loaiza and Rigobon, 2024[7]) model the economic implications of AI on human capabilities that AI cannot reproduce. This literature describes how technology interacts with skills and labour markets but is fundamentally retrospective: it explains how earlier waves of technological change affected the economy, rather than providing a framework for anticipating the future implications of rapidly advancing AI that may move into new areas of capability.

More recent work has tried to estimate more directly which jobs might be affected by AI. (Frey and Osborne, 2017[8]) and (Manyika et al., 2017[9]), were important because they were explicitly prospective and evaluated AI against a full range of work-related abilities. (Eloundou et al., 2023[10]) and (Eisfeldt, Schubert and Zhang, 2023[11]) provide updated measures of AI exposure at the occupational task level, covering the full range of work tasks but focusing only on judgments of the AI capabilities produced by current large language models. (Lassébie and Quintini, 2022[12]) asked experts to assess the automatability, at that particular point in time, of 100 O*NET skills and abilities, and then aggregated these judgements to the occupational level using the importance of each skill or ability within each occupation. (Arquié, Duthoit and Subileau, 2026[13])consider potential occupational task exposure with future AI scenarios. However, none of these studies provides a structure for updating estimates of AI capabilities.

Other approaches use more transparent and updateable signals of AI progress, but without any assessment of the comprehensiveness of AI's capabilities. (Felten, Raj and Seamans, 2019[14]); (Felten, Raj and Seamans, 2021[15]), (Tolan et al., 2021[16]), and (Engberg et al., 2024[17])map AI benchmarks to occupational abilities or tasks; (Kogan et al., 2021[18])and (Webb, 2019[19]) map AI-related patents to occupational tasks; and (Patwardhan et al., 2025[20]) provide performance results on new benchmark tasks drawn from a set of major occupations. Alternatively, some studies look for current rather than potential AI-related activity: (OECD, 2025[21]) and (Alekseeva et al., 2021[22])map AI-related vacancy posting to identify occupations where AI is likely being used; and (Appel et al., 2026[23]) and (Tomlinson et al., 2025[24]) analyse chatbot interactions to identify workplace tasks they are currently being used for. These various studies develop indicators of potential or actual AI use across occupations that could be easily updated, but without providing a comprehensive framework to understand how much of the occupations, abilities, or tasks are beyond the capabilities of the AI systems.

Why we need another measure

Policymakers and business leaders need to understand how AI is likely to affect the economy, the workforce and skill demand over the next 5 to 10 years. That matters for decisions about investment in physical capital, workforce strategy, education and training, and broader economic policy. To be useful for

those purposes, a measure of AI's impact on work should ideally satisfy four requirements at once. It should be forward-looking, so that it says something about technologies that are only beginning to diffuse; transparent, so that users can see how the relevant judgements about future impact are constructed; updateable, so that the analysis can evolve as AI capabilities change and can be used in scenario exercises; and broadly linked to human abilities, so that results can be interpreted in terms of the full range of jobs, tasks and skill demand that are relevant for understanding the implications for human workers.

The measures discussed above have made significant contributions, but none of them satisfies all four requirements. Some are forward-looking but not sufficiently transparent or updateable. Some are transparent and updateable but too narrow to capture the full range of human capabilities relevant to work. Others are well linked to occupations and skills, but remain tied to past technological change or to already-observed use.

The OECD AI Capability Gap Index is designed to bring the four requirements together, based on initial approaches to AI assessment and occupational mapping by (Elliott, 2017[25]), (Elliott, 2017[26]). The novelty of the OECD contribution is therefore not that none of these ideas existed before, but that it attempts to combine them in a single, policy-oriented measure.

3 Using the OECD AI Capability Indicators to Measure AI Exposure

The OECD AI Capability Indicators

The OECD AI Capability Indicators provide a structured, evidence-based framework for assessing what AI systems can and cannot do in comparison with human abilities. Developed by the OECD's AI and Future of Skills (AIFS) project, they cover nine capability domains spanning the main cognitive, social and physical abilities relevant to work, education and everyday life: Language; Social Interaction; Problem Solving; Creativity; Metacognition and Critical Thinking; Knowledge, Learning and Memory; Vision; Manipulation; and Robotic Intelligence.

For the purposes of this paper, the key feature of the indicators is that they are grounded in human psychology and skill theory, which makes it possible to compare AI capabilities directly with the capability requirements of occupations rather than with abstract benchmarks alone. Full level descriptions and worked examples of the scales are provided in Introducing the OECD AI Capability Indicators (OECD, 2025[21]) and the accompanying technical report (OECD, 2025[27]).

The analysis in this paper uses the version of the scales published in June 2025. The indicators will be updated as the evidence base improves and as the OECD's systematic revision process matures, but the results reported here refer to that published version.

How the Indicators can be used to build a Measure of AI Exposure

The OECD AI Capability Indicators provide a straightforward basis for measuring AI exposure. The core idea is simple: If the level of capability that an occupation demands across each domain is known, and the level that AI has currently reached in each domain is also known, the two can be directly compared. Where AI falls short of what an occupation requires, there is a gap. The smaller that gap, the greater the occupation's exposure to AI. In this sense, the measure is best understood as a capability-gap indicator of potential exposure to the current AI frontier. As AI capabilities advance, those gaps will narrow, and the measure updates accordingly.

This new OECD measure is designed to bring together the four requirements identified in Section 2. It is forward-looking by focusing on potential exposure of state-of-the-art capabilities that have not yet been fully applied. It is transparent by separating the assessment of AI capabilities from the assessment of occupational capability demands and making each side of the comparison explicit. It is updateable with each update of the AI Capability Indicators without the need to revise the occupational capability demands. And it is broadly linked to human abilities by being built around a structured capability framework spanning cognitive, social and physical domains.

To operationalise this, human raters and LLMs used the Indicators to assess the capability demands of occupations drawn from O*NET data, placing each occupation on a five-level scale across the nine

capability domains. This builds on the approach pioneered by (Eloundou et al., 2023[10]), but with two important developments. First, raters are given a common grounding in current AI capabilities through the Indicators, rather than relying on their individual knowledge of AI. Second, occupations are assessed across a detailed capability scale rather than assigned to broad categories, enabling more granular analysis of where AI is approaching human-level performance and where significant gaps remain.

The resulting measure, the AI Capability Gap index, expresses AI exposure not as a binary, but as the degree to which AI capabilities are approaching the demands of a given occupation across different domains. This makes it transparent, comparable across occupations, and adaptable as AI continues to develop. It should not be interpreted as a direct prediction of job loss, firm adoption, or the speed of occupational change, all of which depend on factors beyond capability proximity alone.

What analyses can the indicators be used for

Mapping the AI Capability Indicators to occupational requirements and calculating AI Capability Gap indexes can support later analysis of how particular tasks and occupations may evolve as AI capabilities improve. In practice, that kind of downstream analysis would need to combine the exposure measure with explicit assumptions about technological trajectories and with broader social, institutional and policy judgements about where substitution, complementarity or continued human involvement are desirable. (Risi, 2026[28]) provides a first empirical attempt to develop an occupation measure of AI exposure based on this approach and applies it to a labour market analysis in the Italian context. The present paper does not attempt to settle those questions; it establishes the capability-based framework that can support such future work.

4 Manually mapping the AI Capability Indicators to selected occupations

Human ability requirements for occupations can be used to assess AI's implications for work and everyday life by aligning these with the AI Capability Indicators. The capability requirements need to be expressed using the AI Capability Indicators to enable direct comparison with the current and future capabilities of AI systems. This exercise provides a structured link between technological capabilities and the occupational demands that define the world of work today. The approach focuses on the occupation level, offering a high-level yet systematic approximation of how AI could interface with core human competencies across jobs.

The mapping can be conducted using standardised information on occupational characteristics, illustrated in this study by the O*NET database, a comprehensive and internationally recognised data source that covers some 900 US occupations defined in the US Standard Occupational Classification (SOC). O*NET has been widely adopted or adapted by other national labour classification systems, making it a robust foundation for international comparison. Importantly, O*NET includes detailed descriptors of human abilities and skills that are conceptually aligned with the dimensions captured by the AI Capability Indicators. This makes it uniquely suited to bridge AI functionality and job requirements.

This study aims to map each of the approximately 900 occupations from O*NET to the relevant AI Capability Indicators.

The mapping proceeded in four stages. First, a diverse sample of occupations was manually assessed by OECD analysts to create a human benchmark. Second, those ratings were discussed and revised to establish a common reference standard. Third, multiple large language models were employed to rate the same occupations and their assessments were benchmarked against that standard. Fourth, the validated model average was used to extend the ratings to the full O*NET occupational space. Detailed procedures are reported in Annexes 1 to 4.

Selecting a sample of occupations representing the diversity of capability profiles

Because a full manual assessment of roughly 900 occupations was not feasible, the OECD first selected a diverse subset of occupations spanning the main reasoning, social and physical capability profiles in O*NET. The sample was constructed to cover different knowledge domains, occupational groups and job zones, and was refined with support from occupational psychologists to maximise diversity and reduce redundancy. This provided the human-rated benchmark used to anchor the later extension to the full occupational space. Detailed sample-selection procedures are reported in Annex 1.

Individually analysing and rating the demands of the selected occupations

OECD analysts from five directorates then rated the capability demands of the selected occupations across the nine indicators using common background materials and the published level descriptions of the AI Capability Indicators. Individual ratings were aggregated and discussed to create an OECD benchmark for the sample, which served as the reference point for the later LLM-based expansion. The detailed rating protocol and the initial ratings table are reported in Annex 2.

Finding consensus among raters and establishing the OECD standard of truth for the sample of occupations

The consensus process served two functions: it improved consistency in the occupational ratings and revealed where the current indicator descriptions were hardest to apply. The main sources of disagreement concerned occupations with mixed or evolving task bundles, conceptual overlap between some indicators, and differences in how raters used the five-level scales.

Discussion and revision materially improved convergence. The largest gains came in the first round of collective review, especially for the embodied and physical indicators, where disagreement often reflected different assumptions about which tasks should anchor the rating. Later revisions were much smaller and mainly consolidated ratings that had already largely converged.

These discussions also informed the forthcoming revision of the OECD AI Capability Indicators by highlighting ambiguities, overlaps and inconsistencies in the current scale descriptions. Detailed examples of disagreement, the evolution of convergence, and the associated tables are provided in Annex 3.

Establishing the OECD average ratings of capability demands for the sample of occupations

Table 4.1. presents the OECD “standard of truth” as the mean of the most recently revised ratings across the nine AI Capability Indicators for the 40 selected occupations. The occupations are ordered so that the ratings show a clear gradient from occupations with broad and consistently high capability demands to occupations with narrower and more specialised demand profiles. At the high end are occupations such as anaesthesiologists, dentists, elementary school teachers, nursing assistants and childcare workers, which combine strong requirements in language, problem solving, metacognition, and knowledge, learning and memory, often together with substantial social or physical demands. At the low end are occupations such as fast food workers, laundry and dry-cleaning workers, meat cutters and trimmers, and gambling cage workers, where the overall profile is less demanding across most indicators, even though some specific capabilities remain important. This overall ranking is consistent with the logic used to construct the sample, namely to capture the widest possible range of occupational capability profiles rather than only the highest-skill jobs.

Table 4.1. Final OECD mean ratings

| Occupation | Anesthesiologists | | Dentists, General | | Firefighters | | Police Identification and Records Officers | | Elementary School Teachers, Except Special Education | | Nursing Assistants | | Nannies | | Childcare Workers | | Massage Therapists | | Jewelers and Precious Stone and Metal Workers | | Automotive Service Technicians and Mechanics | | Electricians | | Roofers | | Derrick Operators, Oil and Gas | | Excavating and Loading Machine and Dragline Operators | | Crane and Tower Operators | | Bus Drivers, Transit and Intercity | | Meat, Poultry, and Fish Cutters and Trimmers | | Janitors and Cleaners, Except Maids and Housekeeping | | Laundry and Dry Cleaning Workers | | Cargo and Freight Agents | | Fast Food and Counter Workers | | Foresters | | Surveyors | | Chemists | | Geographers | | Airline Pilots, Copilots, and Flight Engineers | | Allergists and Immunologists | | Radiologists | | Fashion Designers | | Video Game Designers | | Tutors | | Marriage and Family Therapists | | Financial Examiners | | Accountants and Auditors | | Concierges | | Retail Salespersons | | Telemarketers | | Court Reporters and Simultaneous Captioners | | Gambling Cage Workers | |
|---|---|---

5 Extending the mapping of AI Capability Indicators to the whole occupational space

This section extends the mapping of AI capability indicators from the pilot sample of 40 occupations to the full occupational space of 879 occupations. The final ratings reported here are based on the average of 20 runs across four large language models: ChatGPT, Mistral, Claude, and Gemini. This ensemble approach was adopted because rating occupational demands at the scale of 879 occupations is not feasible through expert coding alone within a reasonable time frame, while single-model ratings remain vulnerable to model-specific bias, prompt sensitivity, and stochastic variation. By averaging repeated ratings across multiple frontier models, we obtain a more stable and robust estimate of occupational demand profiles.

The decision to rely on LLM-based rating was grounded in the pilot exercise conducted on 40 occupations. That pilot compared model-generated ratings against the OECD standard of truth and showed that the cross-model average performed better than any single model taken on its own. In other words, averaging across ChatGPT, Mistral, Claude, and Gemini reduced idiosyncratic error and produced ratings that were closest to the benchmark. The extension reported in this section therefore builds directly on the validated logic of the pilot: use multiple LLMs, repeat the rating process, and aggregate the outputs.

LLM ratings for occupational groups

The extended mapping reveals a clear structure across the major occupational groups of the SOC (Table 5.1). Cognitive and interaction-intensive capabilities tend to be concentrated in managerial, professional, legal, educational, and social-service occupations, while sensorimotor capabilities are concentrated in manual, technical, and field-based occupations.

Table 5.1. Mean capability demands, by major group

Major Group	Healthcare Practitioners and Technical	Protective Service	Community and Social Service	Legal	Educational Instruction and Library	Management	Business and Financial Operations	Computer and Mathematical	Sales and Related	Life, Physical, and Social Science	Arts, Design, Entertainment, Sports, and Media	Architecture and Engineering	Personal Care and Service	Healthcare Support	Food Preparation and Serving Related	Installation, Maintenance, and Repair	Construction and Extraction	Farming, Fishing, and Forestry	Building and Grounds Cleaning and Maintenance	Transportation and Material Moving	Production	Office and Administrative Support
Major Group Code	**29**	**33**	**21**	**23**	**25**	**11**	**13**	**15**	**41**	**19**	**27**	**17**	**39**	**31**	**35**	**49**	**47**	**45**	**37**	**53**	**51**	**43**
Language	3.5	3.3	4.1	4.2	4.2	3.8	3.7	3.5	3.4	3.7	3.6	3.3	3.0	2.8	2.4	2.2	1.9	2.0	2.4	2.4	1.8	2.8
Social interaction	3.6	3.6	4.3	3.8	4.0	3.8	3.4	2.8	3.5	3.2	3.5	2.8	3.5	3.2	2.9	1.9	1.9	1.9	2.3	2.5	1.3	2.5
Problem solving	3.5	3.7	4.2	3.9	3.9	4.0	3.4	3.3	3.0	3.6	3.4	3.2	3.1	2.6	2.5	2.7	2.3	2.5	2.6	2.6	2.1	2.2
Creativity	2.0	1.7	2.7	2.3	3.2	2.4	2.1	2.3	1.9	2.5	3.3	2.4	1.8	1.3	1.4	1.5	1.2	1.1	1.5	1.0	1.1	1.0
Metacognition	3.5	3.5	3.9	4.0	3.9	4.0	3.5	3.5	2.8	3.7	3.5	3.5	2.7	2.4	2.1	2.7	2.3	2.3	2.5	2.4	2.1	2.2
Knowledge, learning and memory	3.9	3.6	4.0	4.2	4.3	3.9	3.5	3.5	3.0	4.1	3.4	3.6	3.0	3.0	2.7	3.2	2.9	2.8	3.1	2.7	2.3	2.2
Vision	3.2	3.7	0.4	0.3	1.4	1.1	0.9	0.5	1.7	2.2	2.8	2.7	3.1	2.9	3.2	3.4	3.5	3.6	3.5	3.2	3.0	1.1
Manipulation	3.1	2.5	0.2	0.0	1.0	0.5	0.2	0.2	1.2	1.6	1.6	1.5	3.0	3.5	3.3	3.8	3.6	3.2	3.1	2.7	3.4	0.9
Robotic intelligence	2.6	3.2	0.3	0.0	0.9	0.6	0.3	0.2	1.1	1.5	1.4	1.5	3.1	3.2	3.1	3.3	3.5	3.3	3.2	3.0	2.5	0.8
	High demands in all domains		High reasoning demands, incl. social interaction						High social demands	High reasoning and some physical demands			High social and physical demands			High physical demands						Low demands

The first cluster of occupations has relatively high capability demands across all domains and includes healthcare practitioners and protective services. Here, the relevant frontier is not only generative AI, but also embodied and machine-integrated systems that can perceive environments, manipulate objects, and interact with tools, materials, and equipment. The second cluster includes community and social service, legal, education, management, business, computer and mathematical occupations. These groups receive high ratings on language, social interaction (less so for computer and mathematical occupations), and problem solving, as well as metacognition, and knowledge, learning and memory. Sales and related occupations form a distinct intermediate group: they combine relatively high language and social interaction requirements with more moderate reasoning demands and comparatively low physical demands. A middle zone is occupied by groups such as life sciences, arts, and architecture as well as personal care, healthcare support and food preparation and serving occupations. These groups require various combinations of reasoning, social and physical capabilities. Another cluster includes installation and repair, construction, farming, building, transportation, and production occupations. These groups score lower on the language-intensive and reflective indicators, but markedly higher on vision, manipulation, and robotic intelligence. The last group, office and administrative support, is characterised by requiring a broad mix of capabilities, but generally at relatively low levels.

Within-group variation

Several major occupational groups of the SOC contain substantial internal heterogeneity, with notable differences not only between their central tendencies but also in the spread and shape of the underlying occupational distributions. The box plots in Figure 5.1. show that this variation operates at more than one level: in some groups, the interquartile range is wide, indicating that the middle 50% of occupations already differ considerably from one another; in others, the central distribution is more compact, but the whiskers and outliers reveal the presence of more exceptional occupations at the tails. This is particularly relevant for the cognitive and interaction-related indicators such as language, social interaction, problem solving, metacognition, and knowledge, learning and memory. In the strongly professional groups — such as legal, educational, management, and community and social service — the medians are generally high,

confirming that these capabilities are central to the group as a whole. However, the width of the boxes and the reach of the whiskers indicate that these broad occupational families are still internally differentiated. Even where the typical occupation is consistently high on these dimensions, some occupations are markedly more communication-intensive, more knowledge-intensive, or more judgment-intensive than others.

The box plots are especially informative for creativity. Here, the medians identify the groups in which creativity is more central, but the wider interquartile ranges and longer whiskers show that creativity is often unevenly distributed within those groups. In occupational families such as arts and media, and to a lesser extent education and some professional services, the typical range itself is broad. This suggests that creativity is not simply concentrated in a few extreme occupations, but varies meaningfully even across the middle half of occupations within the same group. Where outliers are present, they likely reflect particularly originality-intensive roles at the top end, or more standardized roles at the bottom end.

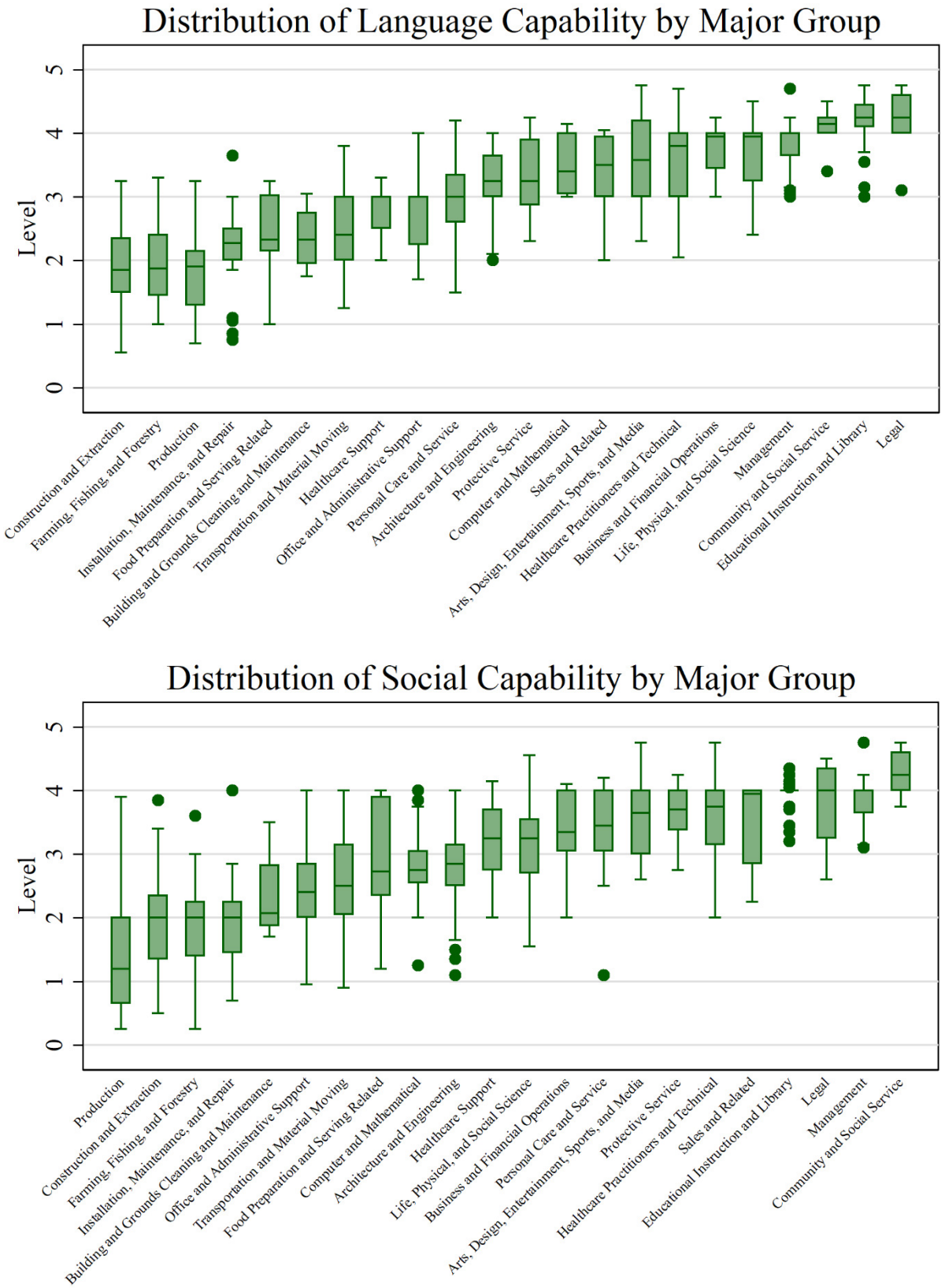
A similar pattern applies to knowledge, learning and memory. High medians in groups such as legal, education, life sciences, and management indicate that knowledge-intensive demands are characteristic of these families overall. But where the boxes and whiskers are wide, this implies that the degree of reliance on formal expertise, accumulated knowledge, and learning capacity still differs considerably across occupations within the same major group. The implication is that these groups are high-knowledge on average, but not uniformly so.

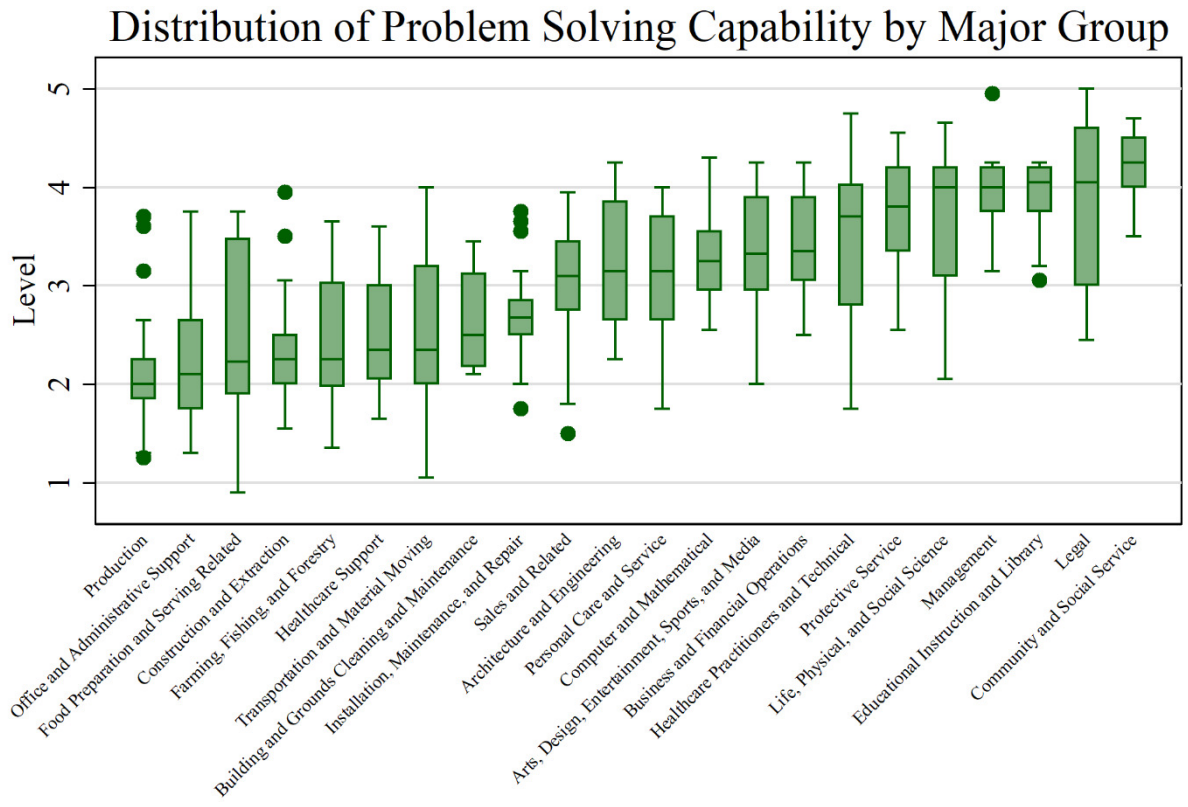
For language and social interaction, the box-plot approach also sharpens the interpretation of service occupations. Where medians are moderate but the interquartile ranges and whiskers are wide, the correct reading is not simply that the group contains a high minimum and a low maximum, but that there is substantial variation even among the central mass of occupations. This is consistent with broad service categories that combine front-line, supervisory, support, clerical, and customer-facing roles with quite different communicative requirements.

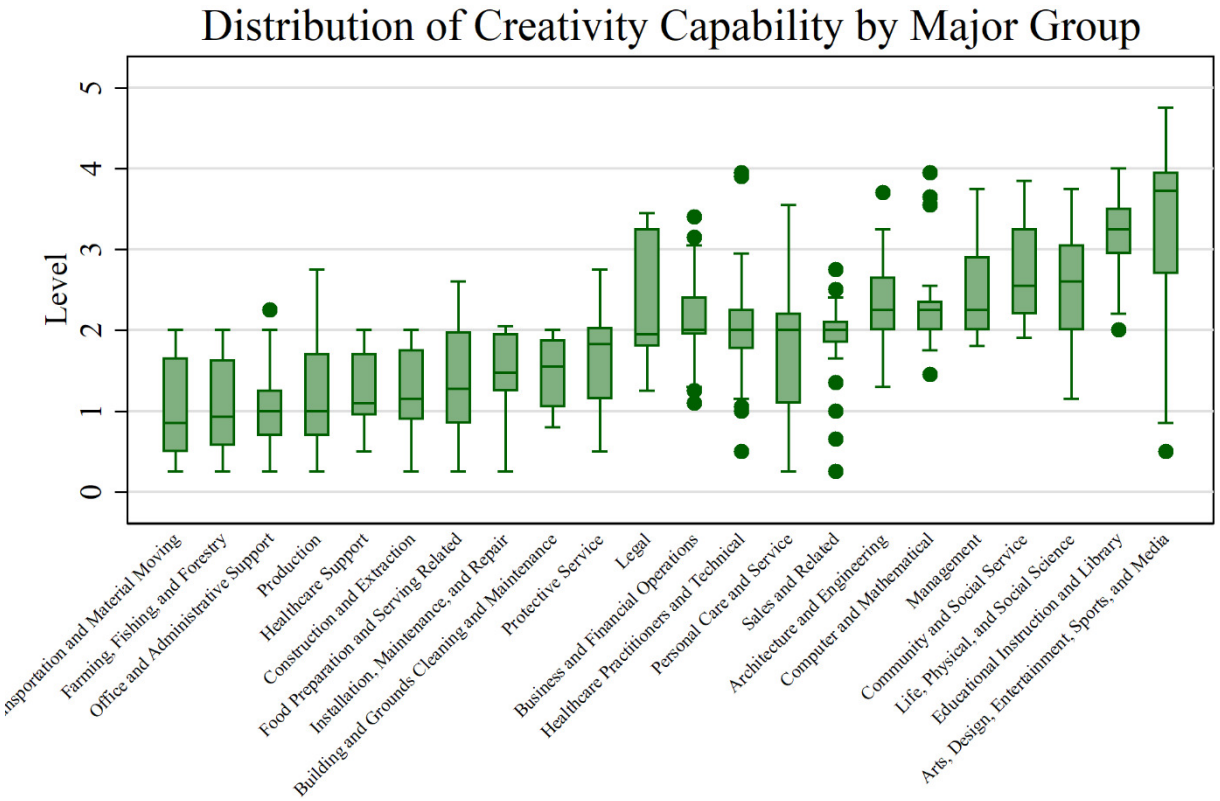
The physical capability indicators — vision, manipulation, and robotic intelligence — show a different type of dispersion. In manual, technical, and field-based occupational groups such as construction, installation, and production among others, the medians are often higher and the boxes are frequently positioned well above zero, indicating that these capabilities are not marginal but characteristic of the group's central occupational profile. When the lower edge of the box and the lower whisker also remain elevated, this suggests that perceptual and object-handling demands are widespread across the group, not confined to a handful of highly specialized roles. In contrast, where cognitively oriented groups such legal, management, computer, business, and community service show boxes compressed near the lower end of the scale for manipulation or robotic intelligence, this indicates that low embodied demand is typical across the bulk of occupations in those groups. Healthcare practitioners and technical stand out here as a highly heterogenous group, combining physical demands at various levels of complexity with high cognitive and interpersonal demands.

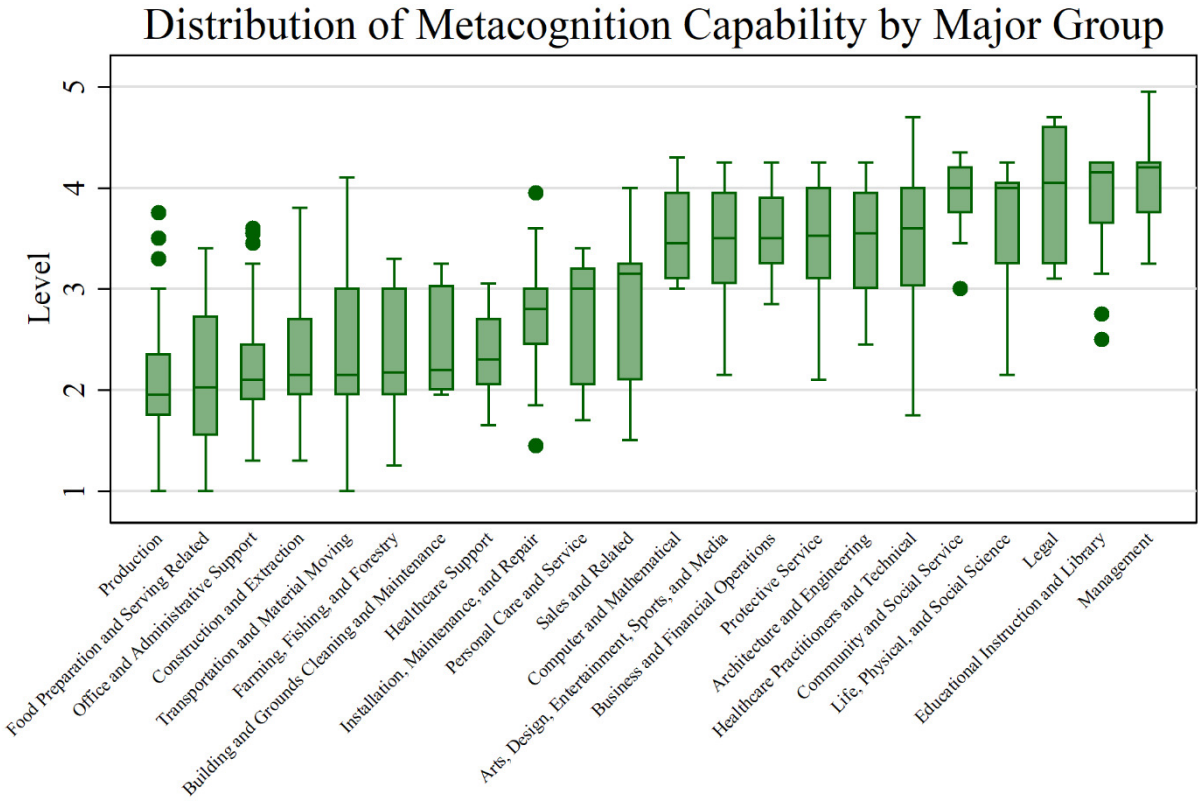
Overall, the box plots reinforce two conclusions. First, the major occupational groups can still be organized into broad clusters with distinct capability-demand profiles. Second, within-group variation remains substantial in many cases, whether because the central distribution is wide or because a small number of occupations lie far from the group norm. This underlines the value of extending the mapping to all 879 occupations rather than relying only on broad group averages.

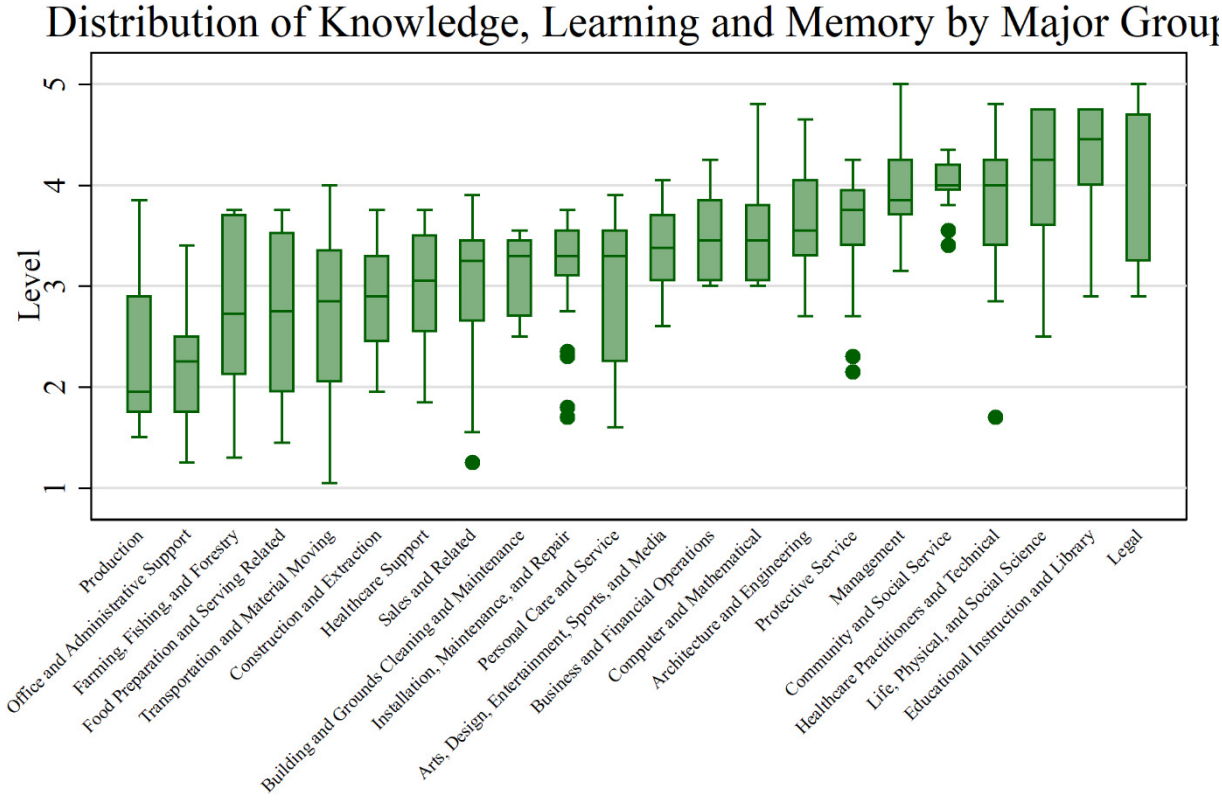
Figure 5.1. Distribution of capability demands within major groups : min, max, median, and interquartile range

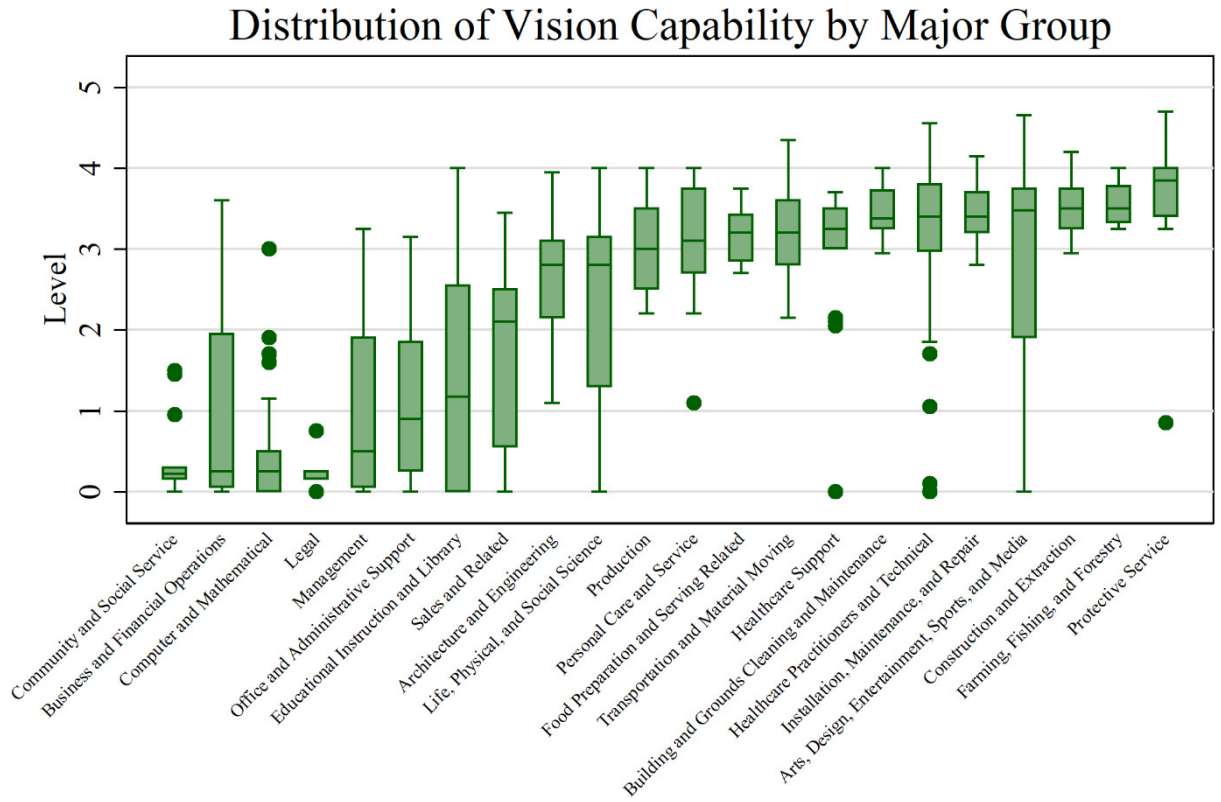


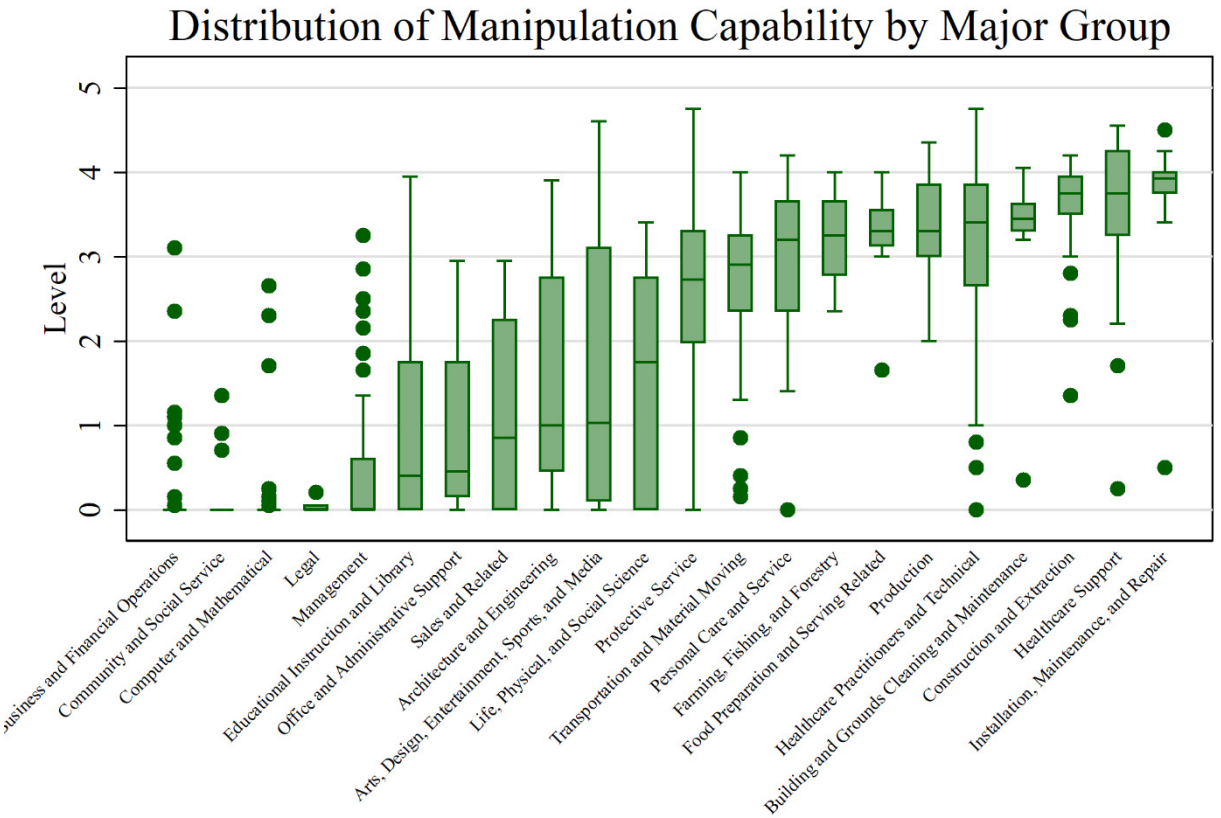

Distribution of Problem Solving Capability by Major Group
Level
1
2
3
4
5
Production
Office and Administrative Support
Food Preparation and Serving Related
Construction and Extraction
Farming, Fishing, and Forestry
Healthcare Support
Transportation and Material Moving
Building and Grounds Cleaning and Maintenance
Installation, Maintenance, and Repair
Sales and Related
Architecture and Engineering
Personal Care and Service
Computer and Mathematical
Arts, Design, Entertainment, Sports, and Media
Business and Financial Operations
Healthcare Practitioners and Technical
Protective Service
Life, Physical, and Social Science
Management
Educational Instruction and Library
Legal
Community and Social Service

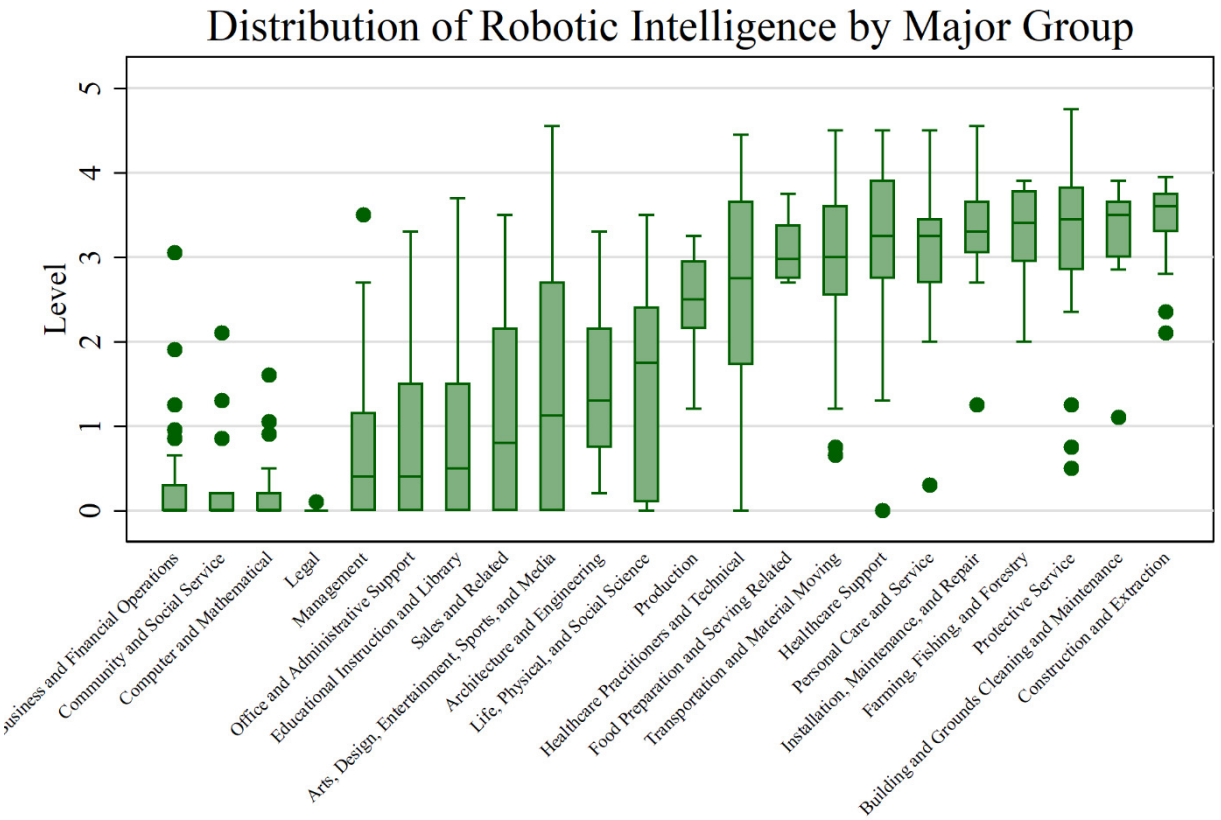

Distribution of Creativity Capability by Major Group
Level
0
1
2
3
4
5
nsportation and Material Moving
Farming, Fishing, and Forestry
Office and Administrative Support
Production
Healthcare Support
Construction and Extraction
Food Preparation and Serving Related
Installation, Maintenance, and Repair
Building and Grounds Cleaning and Maintenance
Protective Service
Legal
Business and Financial Operations
Healthcare Practitioners and Technical
Personal Care and Service
Sales and Related
Architecture and Engineering
Computer and Mathematical
Management
Community and Social Service
Life, Physical, and Social Science
Educational Instruction and Library
Arts, Design, Entertainment, Sports, and Media


Distribution of Metacognition Capability by Major Group
Level
1
2
3
4
5
Production
Food Preparation and Serving Related
Office and Administrative Support
Construction and Extraction
Transportation and Material Moving
Farming, Fishing, and Forestry
Building and Grounds Cleaning and Maintenance
Healthcare Support
Installation, Maintenance, and Repair
Personal Care and Service
Sales and Related
Computer and Mathematical
Arts, Design, Entertainment, Sports, and Media
Business and Financial Operations
Protective Service
Architecture and Engineering
Healthcare Practitioners and Technical
Community and Social Service
Life, Physical, and Social Science
Legal
Educational Instruction and Library
Management


Distribution of Knowledge, Learning and Memory by Major Group
Level
1
2
3
4
5
Production
Office and Administrative Support
Farming, Fishing, and Forestry
Food Preparation and Serving Related
Transportation and Material Moving
Construction and Extraction
Healthcare Support
Sales and Related
Building and Grounds Cleaning and Maintenance
Installation, Maintenance, and Repair
Personal Care and Service
Arts, Design, Entertainment, Sports, and Media
Business and Financial Operations
Computer and Mathematical
Architecture and Engineering
Protective Service
Management
Community and Social Service
Healthcare Practitioners and Technical
Life, Physical, and Social Science
Educational Instruction and Library
Legal


Distribution of Vision Capability by Major Group
Level
0
1
2
3
4
5
Community and Social Service
Business and Financial Operations
Computer and Mathematical
Legal
Management
Office and Administrative Support
Educational Instruction and Library
Sales and Related
Architecture and Engineering
Life, Physical, and Social Science
Production
Personal Care and Service
Food Preparation and Serving Related
Transportation and Material Moving
Healthcare Support
Building and Grounds Cleaning and Maintenance
Healthcare Practitioners and Technical
Installation, Maintenance, and Repair
Arts, Design, Entertainment, Sports, and Media
Construction and Extraction
Farming, Fishing, and Forestry
Protective Service


Distribution of Manipulation Capability by Major Group
Level
0
1
2
3
4
5
Business and Financial Operations
Community and Social Service
Computer and Mathematical
Legal
Management
Educational Instruction and Library
Office and Administrative Support
Sales and Related
Architecture and Engineering
Arts, Design, Entertainment, Sports, and Media
Life, Physical, and Social Science
Protective Service
Transportation and Material Moving
Personal Care and Service
Farming, Fishing, and Forestry
Food Preparation and Serving Related
Production
Healthcare Practitioners and Technical
Building and Grounds Cleaning and Maintenance
Construction and Extraction
Healthcare Support
Installation, Maintenance, and Repair



The extension of the capability mapping to the full occupational space provides a more complete picture of where different forms of AI are likely to matter. Language-based and reasoning-based AI will be most relevant to occupations scoring highly on language, social interaction, problem solving, metacognition, and knowledge-intensive demands. Embodied AI, robotics, and machine vision will matter more in occupations where vision, manipulation, and robotic intelligence are central.

This distinction is important because it cautions against treating AI exposure as a single dimension. The occupational space is not divided simply into high- and low-exposure jobs. Rather, it is differentiated by which capabilities matter most in each occupation. Some occupations are exposed primarily to generative and reasoning systems; others to robotics, machine perception, and physical automation; and some to combinations of the two.

Validating the LLM ratings of capability demands

Because the extended mapping relies on LLM-based ratings to score the occupational demands of 879 occupations, validation is essential for treating the results as a robust empirical foundation. The pilot on 40 occupations already provided strong initial confidence, especially because the average across ChatGPT, Mistral, Claude, and Gemini performed closest to the OECD standard of truth. In addition, the extension to the full occupational space has now been subjected to two further validation steps, which provide additional evidence that the ratings are accurate, stable, and substantively meaningful.

First, occupations were sorted by each indicator and inspected in terms of their rank ordering, including diagonal and cross-indicator consistency checks. This exercise showed that occupations appearing at the top and bottom of each capability dimension generally do so for substantively plausible reasons. Occupations ranked highly on language were concentrated in communication- and knowledge-intensive domains, while occupations ranked highly on manipulation clustered in manual, technical, and operational work. Cross-indicator relationships also followed theoretically expected patterns, such as the association

between language, metacognition, and knowledge demands, and between vision, manipulation, and robotic intelligence. These results provide important evidence that the ratings are not only plausible in aggregate, but also coherent in their detailed ordering.

Second, the results were compared with established occupational patterns in external taxonomies, particularly O*NET. Although the capability indicators used here are not identical to O*NET descriptors, they overlap conceptually with well-established distinctions in the occupational literature. The comparison showed broad alignment with known patterns, notably the contrast between cognitive-communicative occupations and manual-technical occupations. This provides an important form of external validation, suggesting that the LLM-based ratings recover occupational structures that are already well documented in other classification systems rather than generating an artificial or idiosyncratic pattern.

What remains as a further validation step is direct verification by occupational experts, as well as by workers and employers from selected occupations. Even when ratings are statistically coherent and externally aligned with established taxonomies, they still need to be checked for substantive plausibility against lived and professional knowledge of occupational content. Occupational experts can help assess whether the scores correspond to the actual task structure of occupations, identify unexpected outliers, and flag cases where the model may have over- or under-estimated particular capabilities. Input from workers and employers would provide an additional reality check by testing whether the ratings resonate with how occupational demands are understood and experienced in practice. This type of review is likely to be especially valuable for occupations that combine cognitive, social, and physical demands in less straightforward ways.

6 The OECD AI exposure measure

Conceptual framework

Once the capability demands of occupations have been characterised in terms of the AI Capability Indicators, the gap between the level of the capability needed to carry out the job and the current level of AI capabilities can be calculated following the methodology outlined in (Kalamova and Risi, 2025[29]). These capability-specific gaps can then be aggregated at the occupation level to indicate how close current AI capabilities are to the overall capability profile required in each occupation. This should be interpreted as a measure of potential exposure to the current AI frontier. While some parts of a job may offer scope for automation, others may still require human input or human AI collaboration. From this perspective, the OECD AI exposure measure can be understood as a job transformation index. It does not by itself determine whether AI will substitute for human labour, complement workers, or diffuse widely across workplaces.

The AI Capability Gap index can then be further aggregated at industry, regional and country level, offering a flexible metric for assessing the current and potential impact of AI across the economy. Because sectoral structures, workforce skills, firm capabilities, digital infrastructure, and innovation ecosystems differ across places, AI's effects are likely to be uneven geographically as illustrated in Risi (2026). However, this aggregation lies beyond the scope of this paper.

Computing an AI Capability Gap index per capability

Occupations have various capability demands, as expressed in the capability profiles described above. To assess the gap between what a certain job requires and what AI can currently do, the first step is to compute the gap for each required capability. An AI Capability Gap Index per capability, CG_{ji}, is simply the gap between the level of capability *i* needed to carry out a job *j*, H_{ji}, and the current level of AI capability, AI_i: $CG_{ji} = H_{ji} - AI_i$. Here, H refers to the level of human capability demanded by the occupation, and AI refers to the currently assessed level of AI capability in that domain. Since levels range from 1 to 5, the AI Capability Gap index can range between 4 and 0. At level 4, AI is the furthest possible from the required capability level. At level 0, AI has reached the required level of capability, which will be the case for any $AI_i >= H_{ji}$. A zero gap does not imply that AI has already been adopted in the occupation or that the occupation will necessarily be automated; it indicates only that current assessed AI capability is at or above the required level for that domain.

Table 6.1 illustrates the methodology of calculating an AI Capability Gap index per capability with actual numbers. Column H_{ji} provides the level of each of the nine capability domains needed for carrying out an occupation *j*. These levels were assigned following the methodology described in Annex 2 of linking the AI Capability Indicators to the capability demands of occupations and are different for each occupation. The values in the next column AI_i are the current levels of AI capabilities across the nine dimensions and they do not vary by jobs. With these numbers available, the AI Capability Gap index for each of the nine capability indicators *i* can be calculated using the above formula. It turns out that AI systems have already

reached the levels of language, social interaction, creativity, knowledge, learning and memory, and robotic intelligence capabilities needed in this occupation, while others are still far away.

Table 6.1. Computational steps towards an AI Capability Gap index for an example occupation *j*

AI Capability Indicator	H_{ji}	AI_i	CG_{ji}	$Importance_H_{ji}$	CG_j
Language	2	3	0	0.4	
Social interaction	2	2	0	0.3	
Problem solving	3	2	1	0.6	
Creativity	1	3	0	0.3	
Metacognition and critical thinking	3	2	1	0.5	
Knowledge, learning and memory	3	3	0	0.4	
Vision	5	3	2	0.7	
Manipulation	4	2	2	0.8	
Robotic intelligence	2	2	0	0.5	
Total Capability Gap Index CG_j					**4.1**

Computing an occupational total AI Capability Gap index

Occupations require a mix of different capabilities, and AI advances unevenly across these domains. Consequently, the impact of AI on any given job depends on how critical each capability is to that occupation and how well AI performs in each area.

As Table 6.1 shows, AI meets only some of the capability demands, indicating that different job components may be affected to varying degrees. However, drawing firm conclusions about AI's impact on the occupation requires more detailed information. This is especially the case for distinct roles/tasks within the occupation (to be discussed in Next steps) and/or the relative importance of each capability to overall job performance.

To quantify this importance, Table 6.2 outlines a crosswalk for aligning AI Capability Indicators with O*NET's occupational descriptors all of which include measures of importance for specific occupations. Each AI Capability Indicator was matched with one or more relevant O*NET descriptors. Most of these correspond to O*NET's ability scales, although in some cases – such as for social interaction – they draw from O*NET's skill scales instead. For example, the AI Capability Indicator "Language" is associated with O*NET abilities such as Oral Comprehension, Oral Expression, Written Comprehension and Written Expression. Because each AI Capability Indicator corresponds to multiple O*NET constructs, a factor analysis synthesised these values into a single standardised importance score per indicator, reflecting the human performance requirements for each AI Capability Indicator dimension within each occupation. These scores, normalised between 0 and 1, are shown in the column labelled *Importance_Hji* in Table 6.1.

Table 6.2. Crosswalk between AI Capability Indicators and O*NET scales

AI Capability Indicator	Relevant O*NET scales
Language	Oral Comprehension; Oral Expression; Written Comprehension; Written Expression
Social interaction	Co-ordination*; Instructing*; Negotiation*; Persuasion*; Service Orientation*; Social Perceptiveness*
Problem solving	Category Flexibility; Deductive Reasoning; Mathematical Reasoning; Complex Problem Solving*;

AI Capability Indicator	Relevant O*NET scales
	Judgement and Decision Making*; System Analysis*; Systems Evaluation*
Creativity	Fluency of Ideas; Originality
Metacognition and critical thinking	Critical Thinking*; Problem Sensitivity
Knowledge, learning and memory	Memorisation; Inductive Reasoning; Active Learning*; Learning Strategies*
Vision	Depth Perception; Far Vision; Glare Sensitivity; Night Vision; Peripheral Vision; Spatial Orientation
Manipulation	Control Precision; Arm-Hand Steadiness; Finger Dexterity; Manual Dexterity
Robotic intelligence	Response Orientation; Rate Control; Reaction Time; Multilimb Co-ordination; Manual Dexterity; Social Perceptiveness*; Judgement and Decision Making*

Note: The type of O*NET categories is specified as follows: Abilities; Skills*.

To compute a total AI Capability Gap index at the occupational level, each capability-specific AI Capbility Gap index is multiplied by its corresponding importance value, as follows: $CG_{j\square} = \sum_{i=1}^{n}(CG_{j_i} * \text{Importance_}H_{ji\square})$. The outcome is a single composite value that summarises the degree to which current AI capabilities align with the overall requirements of a given occupation. This summary indicator is useful for comparison across occupations, but it necessarily compresses different types of capability gaps into a single value and assumes that the gap between any two capability levels indicates the same incremental difference. Because some capability domains overlap conceptually, the composite index may also partly double-count closely related requirements in some occupations. Occupations with similar total gap indexes may still differ substantially in whether the remaining distance to the AI frontier lies in language, social interaction, metacognition, manipulation or other domains. For these reasons, the total index should be interpreted alongside the underlying domain-specific gaps rather than as a self-sufficient measure of occupational change.

An index value greater than 0 indicates that at least some capability gaps remain between current AI systems and the occupational profile in question. In such cases, AI may still assist with parts of the occupation and create scope for AI human collaboration, but the index does not by itself determine how tasks will be divided in practice. A value of 0 indicates that current assessed AI capability is at or above the required level across the included domains, but this should not on its own be read as implying complete occupational substitution, since actual substitution will also depend on task structure, diffusion and notably the cost of AI and robotics systems (as compared to the cost of human operators), organisational uptake, regulation and other institutional factors.

Since there are nine capability indicators, and the maximum possible value of each capability-specific gap index is 4, the theoretical upper bound of the total gap index is 36, assuming a maximum importance weight of 1 for every capability. In practice, however, this upper bound cannot be reached. Current AI capability already stands above level 1 in every domain, performing at level 2 or 3 across the nine indicators, so the largest gap that can actually arise in a single domain is 2 or 3 rather than 4. Taking current AI levels into account, the effective maximum of the total index is therefore 23, and because occupations rarely assign the maximum importance value of 1 to all capabilities simultaneously, the empirical range of the total AI Capability Gap Index lies well below even this figure. Moreover, equal movements in the total index need not imply equivalent forms of technological change, since progress may occur unevenly across domains and the substantive implications of narrowing a social or metacognitive gap may differ from narrowing a language or perceptual one.

Results

The AI Capability Gap Index shows that the distance between current AI capabilities and occupational requirements varies substantially across both occupations and capability domains. These rankings should be read as indicators of relative capability-based exposure, not as direct predictions of replacement, realised adoption, or the timing of labour-market change. Because the total index compresses multiple domain-specific gaps into a single value, the occupational rankings below should also be read together with the domain-level results rather than as a complete account of how AI may affect each occupation. The occupation-specific examples below are intended to illustrate differences in capability profiles, but not to suggest that any occupation as a whole is either "safe" from AI or likely to be straightforwardly automated. Lower gap values indicate that current AI capabilities are already close to the capability profile required for an occupation and therefore point to relatively high AI exposure. Higher values indicate that AI remains further away from the capabilities required to perform the occupation. In this sense, the index provides an empirical basis for identifying where AI may already affect task performance and where important constraints remain.

At the level of major occupational groups, the results suggest that current AI capabilities are closest to occupations characterised by routine information-processing and administrative tasks (Table 6.3). Office and Administrative Support Occupations record the lowest total gap index, at 0.8. They are followed by the groups of Production Occupations, at 1.8, Food Preparation and Serving Related Occupations, at 2.4, and Sales and Related Occupations, at 2.6. By contrast, the highest total gap indexes are observed in Healthcare Practitioners and Technical and Protection Occupations (5.6 and 5.2, correspondingly), in Community and Social Service (6.4), Legal (5.8), Educational Instruction (5.8) and Management Occupations (5.5), followed by Science (4.8), Arts, Sports and Media (4.1), and Installation, Maintenance and Repair (3.5) occupations. At the same time, these group-level averages mask substantial variation within many occupational groups, with some occupations showing zero capability gaps while others still have gaps of several levels across multiple domains. Overall, these patterns suggest that AI exposure is currently highest in occupations with standardised, codifiable and repetitive task content, including routine inspection and monitoring tasks, and lower in occupations that rely more heavily on dexterity, judgement, social interaction, interpretation and responsibility.

Table 6.3. Gap between occupational capability demands and current AI capabilities and total AI Capability Gap index, by major occupational groups

Major Group	Healthcare Practitioners and Technical	Protective Service	Community and Social Service	Legal	Educational Instruction and Library	Management	Business and Financial Operations	Computer and Mathematical	Sales and Related	Life, Physical, and Social Science	Arts, Design, Entertainment, Sports, and Media	Architecture and Engineering	Personal Care and Service	Healthcare Support	Food Preparation and Serving Related	Installation, Maintenance, and Repair	Construction and Extraction	Farming, Fishing, and Forestry	Building and Grounds Cleaning and Maintenance	Transportation and Material Moving	Production	Office and Administrative Support	
Major Group Code	**29**	**33**	**21**	**23**	**25**	**11**	**13**	**15**	**41**	**19**	**27**	**17**	**39**	**31**	**35**	**49**	**47**	**45**	**37**	**53**	**51**	**43**	
Language	0.6	0.4	1.1	1.2	1.2	0.8	0.7	0.5	0.5	0.7	0.6	0.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	**0.4**
Social interaction	1.6	1.6	2.3	1.8	2.0	1.8	1.4	0.8	1.5	1.2	1.5	0.8	1.5	1.2	1.0	0.2	0.2	0.3	0.4	0.6	0.1	0.5	**1.1**
Problem solving	1.5	1.7	2.2	1.9	1.9	2.0	1.4	1.3	1.0	1.6	1.4	1.2	1.1	0.6	0.7	0.7	0.3	0.5	0.6	0.6	0.2	0.4	**1.1**
Creativity	0.0	0.0	0.1	0.1	0.3	0.0	0.0	0.1	0.0	0.1	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.1**
Metacognition	1.5	1.5	1.9	2.0	1.9	2.0	1.5	1.5	0.8	1.7	1.5	1.5	0.7	0.4	0.4	0.7	0.4	0.4	0.5	0.5	0.2	0.3	**1.1**
Knowledge, learning and memory	0.9	0.7	1.0	1.3	1.3	0.9	0.5	0.5	0.3	1.1	0.4	0.6	0.3	0.2	0.2	0.3	0.2	0.3	0.3	0.2	0.1	0.0	**0.5**
Vision	0.5	0.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.5	0.1	0.3	0.2	0.2	0.4	0.5	0.6	0.5	0.4	0.3	0.0	**0.2**
Manipulation	1.3	0.8	0.0	0.0	0.1	0.1	0.0	0.0	0.2	0.4	0.5	0.4	1.1	1.6	1.3	1.8	1.6	1.2	1.4	0.8	1.4	0.1	**0.7**
Robotic intelligence	0.9	1.4	0.0	0.0	0.2	0.0	0.0	0.0	0.2	0.3	0.3	0.2	1.1	1.3	1.1	1.3	1.5	1.3	1.3	1.1	0.5	0.1	**0.6**
Overall AI Capability Gap Index	**5.6**	**5.2**	**6.4**	**5.8**	**5.8**	**5.5**	**3.7**	**3.1**	**2.6**	**4.8**	**4.1**	**3.4**	**3.0**	**2.8**	**2.4**	**3.5**	**3.1**	**2.7**	**2.7**	**2.8**	**1.8**	**0.8**	

The results also vary substantially across capability domains. On average across occupational groups, the largest gaps are observed in social interaction, problem solving and metacognition, all three at 1.1. These are followed by the physical domains of manipulation (0.7) and robotic intelligence (0.6), which may prove the most stubborn to close. By contrast, the average gaps are much smaller in vision, at 0.2, language, at 0.4, and knowledge, learning and memory, at 0.5. Creativity records the smallest average gap, at 0.1. (This result should be interpreted with particular care, since level 5 of the creativity scale captures exceptional or world-class creativity that is rarely required of whole occupations, effectively compressing the active range of occupational creativity ratings and contributing to a smaller measured gap compared to the other capability scales.) These findings suggest that current AI capabilities are closest to occupational requirements involving language processing, codified knowledge, visual perception and some forms of content generation, while they remain further from capabilities related to dexterity, contextual reasoning, social understanding, and complex judgement.

The occupational rankings within each capability domain provide additional insight into these patterns (Table 6.4a.). In the language domain, the largest gaps are found in occupations such as judges, English language and literature teachers, foreign language and literature teachers, poets and lyricists, and interpreters and translators. In social interaction, the least exposed occupations include chief executives, mental health counsellors, clergy, marriage and family therapists, psychiatrists and actors. In problem solving, among the occupations with the largest gaps are judges, chief executives, psychiatrists, mental health counsellors, clinical and counselling psychologists, and marriage and family therapists. These occupations are characterised by a strong need for context-dependent interpretation, interpersonal judgement and responsibility, which remain difficult to replicate with current AI systems.

A similar pattern is observed in the domains linked to higher-order cognition. In metacognition and critical thinking, the largest gaps are observed for chief executives, judges, psychiatrists, lawyers, and mental health counsellors. In knowledge, learning and memory, occupations such as chief executives, judges, family medicine physicians, psychiatrists and scientists rank among the least exposed. These findings suggest that, even where AI systems perform well on narrower tasks, many occupations continue to rely on the integration of knowledge, reasoning and judgement in ways that remain beyond current frontier capabilities. In occupations such as chief executives or judges, this should be understood as reflecting the breadth of the occupational role as currently structured—including contextual judgement, responsibility and dependence on wider institutional or collaborative processes—rather than as a claim that individual incumbents are broadly beyond AI assistance or substitution.

The physical capability domains highlight a different set of occupations. In vision, the largest gaps are observed for firefighters, athletes, ophthalmologists, lifeguards, and referees. In manipulation, the least exposed occupations include firefighters, oral and maxillofacial surgeons, obstetricians and gynaecologists, surgical assistants and urologists. In robotic intelligence, the occupations with the largest gaps include firefighters, commercial divers, police officers, athletes, personal care aides, home health aides, childcare workers, and bus drivers. These occupations involve embodiment, dexterity, physical adaptability and action in dynamic environments, all of which continue to limit the ability of current AI systems to perform the relevant tasks.

Many occupations have capability-specific gaps equal to zero, meaning that current AI capabilities already meet or exceed the occupational requirement for the relevant capability. However, the capability domain in question is not always central to the occupation, so a zero gap does not necessarily imply high AI exposure in practice. To identify the occupations that are most exposed in a more meaningful sense, Table 6.4b lists occupations with zero capability gaps that also have the highest importance values for the relevant capabilities.

Table 6.4a. Top 10 occupations with the lowest AI exposure (largest AI Capability Gap index), by capability domain

Language			Social Interaction			Problem Solving		
Group	Occupation Title	Gap	Group	Occupation Title	Gap	Group	Occupation Title	Gap
23	Judges, Magistrate Judges, and Magistra	1.8	11	Chief Executives	2.8	23	Judges, Magistrate Judges, and Magistrates	3
25	Foreign Language and Literature Teache	1.8	21	Marriage and Family Therapists	2.8	11	Chief Executives	2.9
25	English Language and Literature Teacher	1.8	29	Psychiatrists	2.8	29	Psychiatrists	2.8
27	Poets, Lyricists and Creative Writers	1.8	21	Mental Health Counselors	2.8	21	Mental Health Counselors	2.7
27	Interpreters and Translators	1.8	27	Actors	2.8	21	Marriage and Family Therapists	2.7
29	Psychiatrists	1.7	21	Clergy	2.7	19	Clinical and Counseling Psychologists	2.7
25	Philosophy and Religion Teachers, Posts	1.7	21	Mental Health and Substance Abuse Sc	2.6	23	Lawyers	2.6
11	Chief Executives	1.7	19	Clinical and Counseling Psychologists	2.6	21	Mental Health and Substance Abuse Social	2.6
23	Lawyers	1.6	23	Arbitrators, Mediators, and Conciliator	2.5	33	Detectives and Criminal Investigators	2.6
25	Law Teachers, Postsecondary	1.6	21	Child, Family, and School Social Worke	2.3	21	Clergy	2.5
Creativity			**Metacognition**			**Knowledge, Learning and Memory**		
Group	Occupation Title	Gap	Group	Occupation Title	Gap	Group	Occupation Title	Gap
27	Poets, Lyricists and Creative Writers	1.8	11	Chief Executives	2.9	23	Judges, Magistrate Judges, and Magistrates	2
27	Fine Artists, Including Painters, Sculptor	1.4	29	Psychiatrists	2.7	11	Chief Executives	2
27	Choreographers	1.3	23	Judges, Magistrate Judges, and Magistr	2.7	15	Computer and Information Research Scien	1.8
27	Actors	1.3	23	Lawyers	2.6	29	Family Medicine Physicians	1.8
27	Music Directors and Composers	1.3	21	Mental Health Counselors	2.3	25	Sociology Teachers, Postsecondary	1.8
27	Musicians and Singers	1.1	15	Computer and Information Research S	2.3	19	Bioinformatics Scientists	1.8
27	Producers and Directors	1.1	27	Art Directors	2.3	25	Environmental Science Teachers, Postsecd	1.8
27	Fashion Designers	1.1	25	Special Education Teachers, Middle Sc	2.3	29	Psychiatrists	1.8
25	Art, Drama, and Music Teachers, Postse	1	11	Computer and Information Systems Ma	2.3	19	Biologists	1.8
25	English Language and Literature Teacher	1	11	Education and Childcare Administrator	2.3	19	Molecular and Cellular Biologists	1.8
Vision			**Manipulation**			**Robotic Intelligence**		
Group	Occupation Title	Gap	Group	Occupation Title	Gap	Group	Occupation Title	Gap
33	Firefighters	1.7	29	Surgical Assistants	2.8	33	Police and Sheriff's Patrol Officers	2.8
27	Athletes and Sports Competitors	1.7	33	Firefighters	2.8	33	Firefighters	2.8
29	Ophthalmologists, Except Pediatric	1.6	29	Obstetricians and Gynecologists	2.7	27	Athletes and Sports Competitors	2.6
33	Lifeguards, Ski Patrol, and Other Recreat	1.5	29	Urologists	2.7	49	Commercial Divers	2.6
27	Umpires, Referees, and Other Sports Off	1.4	29	Oral and Maxillofacial Surgeons	2.6	53	Bus Drivers, Transit and Intercity	2.5
53	Bus Drivers, Transit and Intercity	1.3	29	Ophthalmologists, Except Pediatric	2.6	31	Personal Care Aides	2.5
53	Commercial Pilots	1.3	27	Athletes and Sports Competitors	2.6	39	Childcare Workers	2.5
53	Captains, Mates, and Pilots of Water Ves	1.3	31	Personal Care Aides	2.6	31	Home Health Aides	2.5
29	Oral and Maxillofacial Surgeons	1.3	29	Dentists, General	2.5	29	Surgical Assistants	2.4
33	Police and Sheriff's Patrol Officers	1.3	49	Commercial Divers	2.5	33	Fish and Game Wardens	2.4

Table 6.4b. Top 10 occupations with the highest AI exposure (smallest AI Capability Gap index), by capability domain

Language		Social Interaction		Problem Solving	
Group	Occupation Title	Group	Occupation Title	Group	Occupation Title
31	Medical Transcriptionists	49	Manufactured Building and Mobile Home I	43	Statistical Assistants
47	Energy Auditors	49	Bicycle Repairers	51	Cooling and Freezing Equipment Operator
19	Remote Sensing Scientists and Techno	51	Biofuels Processing Technicians	51	Rolling Machine Setters, Operators, and T
43	Secretaries and Administrative Assistar	47	Stonemasons	51	Photographic Process Workers and Proces
17	Industrial Engineers	13	Logistics Analysts	53	Conveyor Operators and Tenders
17	Geodetic Surveyors	47	Hazardous Materials Removal Workers	51	Crushing, Grinding, and Polishing Machin
43	Loan Interviewers and Clerks	49	Mechanical Door Repairers	29	Pharmacy Technicians
17	Mechatronics Engineers	51	Photographic Process Workers and Proces	53	Hoist and Winch Operators
19	Environmental Science and Protection	19	Biochemists and Biophysicists	45	Log Graders and Scalers
29	Medical Dosimetrists	49	Musical Instrument Repairers and Tuners	43	Weighers, Measurers, Checkers, and Sam
Creativity		**Metacognition**		**Knowledge, Learning and Memory**	
Group	Occupation Title	Group	Occupation Title	Group	Occupation Title
19	Epidemiologists	53	Hoist and Winch Operators	43	Statistical Assistants
17	Automotive Engineers	53	Gas Compressor and Gas Pumping Station	15	Clinical Data Managers
17	Robotics Engineers	31	Medical Equipment Preparers	19	Precision Agriculture Technicians
15	Health Informatics Specialists	51	Cooling and Freezing Equipment Operators	19	Biological Technicians
13	Business Continuity Planners	49	Bicycle Repairers	13	Credit Analysts
29	Preventive Medicine Physicians	47	Roof Bolters, Mining	29	Neurodiagnostic Technologists
19	Transportation Planners	51	Rolling Machine Setters, Operators, and Te	15	Document Management Specialists
11	Emergency Management Directors	47	Stonemasons	17	Architectural and Civil Drafters
19	Bioinformatics Scientists	29	Pharmacy Technicians	31	Medical Assistants
21	Mental Health and Substance Abuse Sc	53	Pump Operators, Except Wellhead Pumper	49	Telecommunications Equipment Installer
Vision		**Manipulation**		**Robotic Intelligence**	
Group	Occupation Title	Group	Occupation Title	Group	Occupation Title
53	Tank Car, Truck, and Ship Loaders	53	Subway and Streetcar Operators	51	Crushing, Grinding, and Polishing Machin
53	Dredge Operators	53	Captains, Mates, and Pilots of Water Vesse	51	Adhesive Bonding Machine Operators and
53	Airfield Operations Specialists	53	Locomotive Engineers	51	Textile Winding, Twisting, and Drawing Ou
13	Government Property Inspectors and Ir	47	Solar Energy Installation Managers	51	Extruding and Forming Machine Setters, C
51	Biofuels Processing Technicians	49	First-Line Supervisors of Mechanics, Instal	53	Machine Feeders and Offbearers
53	Conveyor Operators and Tenders	19	Precision Agriculture Technicians	51	Shoe Machine Operators and Tenders
43	Meter Readers, Utilities	33	First-Line Supervisors of Police and Detect	51	Packaging and Filling Machine Operators a
51	Hydroelectric Plant Technicians	11	Food Service Managers	39	Motion Picture Projectionists
51	Petroleum Pump System Operators, Re	29	Clinical Nurse Specialists	51	Computer Numerically Controlled Tool Pr
53	Aircraft Cargo Handling Supervisors	17	Electronics Engineers, Except Computer	51	Cleaning, Washing, and Metal Pickling Equ

In language and social interaction, the most exposed occupations point to tasks that rely on structured communication, documentation, information exchange and routine interaction. In the language domain, exposure is concentrated in activities such as transcribing, recording, summarising, classifying and communicating information, as well as interpreting technical or administrative material. In social interaction, the zero-gap occupations suggest exposure in forms of interaction that are relatively bounded, practical or transactional, such as coordinating with clients or co-workers, explaining procedures, gathering information, or supporting service delivery. These results suggest that current AI capabilities are closest to occupational requirements where language and interaction are structured, information-based and relatively standardised, rather than where they require complex persuasion, emotional judgement or responsibility for sensitive interpersonal outcomes.

In the reasoning-related domains—problem solving, creativity, metacognition, and knowledge, learning and memory—the most exposed occupations highlight activities based on codified knowledge, structured analysis and procedural decision-making. Exposure is particularly apparent in tasks involving data handling, recordkeeping, monitoring, classification, technical documentation, planning within established frameworks, and applying specialised knowledge to well-defined problems. The results also suggest exposure in some forms of applied creativity, such as generating, adapting or combining technical and informational content, rather than open-ended or exceptional creativity. Overall, current AI capabilities appear closest to reasoning requirements that are rule-based, evidence-based, data-rich or supported by

established procedures, while larger constraints remain where tasks require open-ended judgement, accountability, contextual interpretation or integration of competing objectives.

In the physical domains—vision, manipulation and robotic intelligence—the most exposed occupations point to activities performed in relatively structured, equipment-mediated or controlled environments. In vision, exposure appears strongest for inspection, monitoring and gauging tasks and for the visual guidance of equipment and material movement, especially where objects, settings or procedures are relatively standardised. In manipulation, the exposed activities often involve operating, guiding or supervising equipment rather than highly dexterous manual work in unpredictable settings. In robotic intelligence, exposure is concentrated in repetitive machine-operation and production processes, where tasks follow regular sequences and take place in organised environments. These patterns suggest that current AI and robotic capabilities are closest to physical requirements involving structured perception, routine equipment use and repeatable processes, while remaining more limited in tasks requiring fine dexterity, physical adaptability, mobility in complex environments or real-time responses to unpredictable conditions.

The ranking of occupations by total AI Capability Gap index confirms these broader patterns (Table 6.5). The occupations with the highest AI exposure are predominantly administrative. They include word processors and typists, file clerks, data entry keyers, bookkeeping, accounting and auditing clerks, billing and posting clerks, weighers, measurers, checkers and samplers, payroll and timekeeping clerks, shipping, receiving and inventory clerks, statistical assistants and medical transcriptionists, all with gap indexes equal to or close to zero. By contrast, the occupations with the lowest overall exposure include, chief executives, ophthalmologists, firefighters, obstetricians and gynaecologists, police and sheriff's patrol officers, urologists, psychiatrists, lawyers, judges, and anaesthesiologists. These rankings indicate that the full capability profiles of these occupations still include domains where substantial gaps remain; they should not be interpreted as meaning that such occupations are immune to AI-driven change or that many component tasks within them are not already being affected. These occupations combine several of the domains in which AI remains furthest from human performance, including social interaction, problem solving, metacognition and, in many cases, advanced physical capability.

Table 6.5. Occupations with the highest versus lowest AI exposure overall, by AI Capability Gap index

Highest AI exposure			Lowest AI exposure		
Group	Occupation Title	Total AI Capability Gap Index	Group	Occupation Title	Total AI Capability Gap Index
43	Billing and Posting Clerks	0	11	Chief Executives	11.71
43	Word Processors and Typists	0	29	Ophthalmologists, Except Pediatric	10.56
43	File Clerks	0	29	Obstetricians and Gynecologists	10.41
43	Bookkeeping, Accounting, and Auditing Clerk	0	33	Firefighters	10.01
43	Data Entry Keyers	0	29	Urologists	9.86
43	Weighers, Measurers, Checkers, and Sample	0	29	Psychiatrists	9.76
43	Payroll and Timekeeping Clerks	0.03	33	Police and Sheriff's Patrol Officers	9.54
43	Shipping, Receiving, and Inventory Clerks	0.06	23	Lawyers	9.53
43	Statistical Assistants	0.08	23	Judges, Magistrate Judges, and Magistrates	9.5
31	Medical Transcriptionists	0.09	29	Anesthesiologists	9.4

The total AI Capability Gap index results suggest that the near-term impact of AI is likely to be concentrated in occupations whose task content is highly structured and codifiable, rather than in occupations simply classified as high- or low-skill. Conversely, the occupations with the lowest exposure are typically those that combine multiple capability requirements that current AI systems do not yet satisfy. The gap index therefore provides a more differentiated picture of potential AI impacts across the labour market. It

highlights that AI exposure depends on the specific configuration of capabilities required by each occupation, and that substantial distance remains between current AI systems and human work in many domains that are central to decision making, care, leadership and skilled professional practice. For that reason, the total index is most useful as a summary of overall exposure patterns, while the domain-specific results remain essential for understanding the particular forms of capability distance that drive those rankings.

For use in further empirical work, the OECD AI Capability Gap Index was also reversed so that higher values indicate higher AI exposure, rather than larger remaining capability gaps. The resulting exposure index was then normalised, to support different types of economic analysis, including comparisons across occupations, sectors, workers or countries. To facilitate such use, the OECD has made the full dataset available at Introducing the OECD AI Capability Indicators - OECD.AI. This dataset includes the total occupational AI Capability Gap index, the reversed index, the capability-specific gap indexes and the underlying occupational capability demand ratings.

Comparison with Felten et al. and the implications of broader capability coverage

(Felten, Raj and Seamans, 2021[15]) provide one of the most widely used occupation-level measures of AI exposure. Their approach links progress in AI applications to occupational abilities by first identifying AI capabilities for which benchmark evidence is available and then assessing the extent to which these capabilities correspond to the abilities required in different occupations. The resulting measure captures the degree to which advances in AI are likely to affect occupations, based on the overlap between AI-relevant capabilities and occupational skill requirements. It is therefore a useful benchmark against which to compare the OECD AI Capability Gap index, particularly because both approaches seek to assess potential exposure through the capability content of jobs.

One important difference between the OECD measure and (Felten, Raj and Seamans, 2021[15]), however, is the breadth of capability coverage on which each is based. Felten et al. links occupational exposure to a narrower set of AI functions centred on language, information processing and related benchmarked capabilities, whereas the OECD index compares occupational requirements with a broader capability frontier that also includes social, sensory and embodied domains.

This difference in scope matters because occupations rarely rely on only one kind of capability. A measure centred mainly on language and information-processing capabilities will overlap most with the OECD framework in routine clerical and administrative work but can diverge more sharply in occupations where social interaction, perception, manipulation or robotic intelligence are central.

The comparison of the two measures for a sample of 610 occupations matched between the OECD and Felten et al. datasets confirms that they are related but not interchangeable (Table 6.6). Correlations are strongest for language, problem solving, metacognition, social interaction, and knowledge, learning and memory, but they are negative for the more embodied domains of vision, manipulation and robotic intelligence. The key implication is that the two measures overlap most in occupations driven by language and information-processing capabilities, while diverging more where social, perceptual and embodied capabilities play a larger role.

Table 6.6. Correlations between Felten et al. measure and the OECD AI Capability Gap indexes

	Correlation with OECD AI Capability Gap Index	Correlation with OECD capability demands
Language	0.70	0.82
Social interaction	0.63	0.63
Problem solving	0.67	0.66
Creativity	0.27	0.61
Metacognition	0.71	0.71
Knowledge, learning and memory	0.58	0.56
Vision	-0.45	-0.72
Manipulation	-0.78	-0.84
Robotic intelligence	-0.67	-0.81
Total AI Capability Gap Index	0.40	

These patterns are consistent with the construction of Felten et al.'s measure. In practice, it behaves mainly as an indicator of occupational exposure to the subset of AI capabilities associated with language, information processing and reasoning-related functions. The OECD measure, by contrast, is designed to compare occupational requirements with a broader capability frontier that includes not only cognitive but also social, sensory and embodied domains.

The contrast becomes especially clear when comparing the occupations that appear near the top and bottom of the two rankings (Table 6.7). Occupations such as judges, genetic counsellors, actuaries and mathematicians rank as highly exposed in Felten et al. because they rely heavily on the kinds of cognitive capabilities around which that measure is built. In the OECD measure, however, they rank much lower because their full occupational profiles also depend on capabilities—such as social judgement, context-sensitive interpretation, responsibility, or embodied decision-making—that current AI still does not fully match.

Table 6.7. Occupations with the highest versus lowest AI exposure, by Felten et al measure and the OECD AI Capability Gap index

A. 10 most exposed occupations by Felten et al. as compared to OECD exposure ranking
1=most exposed, 610=least exposed

Code	Occupation title	Felten et al.	OECD
29	Genetic Counselors	1	497
13	Financial Examiners	2	443
15	Actuaries	3	421
13	Purchasing Agents, Except Wholesale, Retail, and Farm Products	4	378
13	Budget Analysts	5	261
23	Judges, Magistrate Judges, and Magistrates	6	605
43	Procurement Clerks	7	37
13	Accountants and Auditors	8	328
15	Mathematicians	9	483
23	Judicial Law Clerks	10	426

B. 10 least exposed occupations by Felten et al. as compared to OECD exposure ranking
1=most exposed, 610=least exposed

Code	Occupation title	Felten et al.	OECD
27	Dancers	610	313
47	Helpers--Painters, Paperhangers, Plasterers, and Stucco Masons	609	135
47	Reinforcing Iron and Rebar Workers	608	288
51	Pressers, Textile, Garment, and Related Materials	607	151
47	Helpers--Brickmasons, Blockmasons, Stonemasons, and Tile and Marble Setters	606	167
35	Dining Room and Cafeteria Attendants and Bartender Helpers	605	128
47	Fence Erectors	604	305
47	Helpers--Roofers	603	213
51	Slaughterers and Meat Packers	602	217
37	Landscaping and Groundskeeping Workers	601	221

C. 10 most exposed occupations by OECD as compared to Felten et al. exposure ranking
1=most exposed, 610=least exposed

Code	Occupation title	Felten et al.	OECD
43	Bookkeeping, Accounting, and Auditing Clerks	150	1
43	Word Processors and Typists	213	2
43	Data Entry Keyers	243	3
43	File Clerks	247	4
43	Weighers, Measurers, Checkers, and Samplers, Recordkeeping	314	5
43	Payroll and Timekeeping Clerks	102	6
43	Statistical Assistants	32	7
31	Medical Transcriptionists	145	8
43	Switchboard Operators, Including Answering Service	166	9
43	Proofreaders and Copy Markers	86	10

D. 10 least exposed occupations by OECD as compared to Felten et al. exposure ranking
1=most exposed, 610=least exposed

Code	Occupation title	Felten et al.	OECD
11	Chief Executives	63	610
29	Obstetricians and Gynecologists	249	609
29	Psychiatrists	92	608
33	Police and Sheriff's Patrol Officers	394	607
23	Lawyers	67	606
23	Judges, Magistrate Judges, and Magistrates	6	605
29	Anesthesiologists	252	604
29	Oral and Maxillofacial Surgeons	321	603
21	Mental Health Counselors	57	602
53	Airline Pilots, Copilots, and Flight Engineers	317	601

The reverse pattern appears for many manual and embodied occupations. Jobs that rank among the least exposed in Felten et al., such as roofers, brickmasons, or landscaping workers, are often more exposed in the OECD framework once machine vision, manipulation and robotic intelligence are brought into view. At the same time, the overlap between the two measures is strongest in clerical and routine information-processing work, where both identify occupations such as statistical assistants, payroll and timekeeping clerks, proofreaders, and bookkeeping clerks as highly exposed.

The occupations ranked as least exposed in the OECD measure should also be interpreted with care. Cases such as chief executives, judges, lawyers or psychiatrists should not be read as indicating that these occupations are broadly insulated from AI, but rather that their full capability profiles include domains that remain difficult for current systems to match. The ranking therefore reflects the breadth of occupational requirements, not a claim that such jobs are 'safe' from technological change.

Taken together, the comparison shows that the two measures answer related but distinct questions. Felten et al. is most informative for a narrower subset of AI capabilities centred on language and information processing, while the OECD framework is designed to assess exposure relative to a broader capability frontier. The choice between them therefore depends on the analytical question at hand.

Seen in that light, the OECD contribution is not that earlier measures were wrong, but that broader capability coverage changes the occupational picture in substantively important ways. That matters especially for policy, because it shifts attention from a one-dimensional view of exposure toward a more differentiated account of how AI may affect occupations that combine cognitive, social and embodied demands.

7 Scenario applications and future directions

Scenario-based application: A case study of increasing AI cognitive capabilities by one level

One reason the OECD AI exposure measure is forward-looking is that, because it is built from capability-specific gaps, it can be recalculated under explicit scenarios about future AI progress. Most available indicators provide a static picture, linking current AI applications to current job tasks. By contrast, the AI Capability Gap index measures the distance between occupational capability requirements and the AI frontier, making it possible to update the assessment as AI capabilities evolve or under future scenarios of AI advancement. This makes the indicator particularly well suited to scenario analysis. Because the index is built from capability-specific gaps, it can be recalculated under alternative assumptions about future progress in different domains, such as language, problem solving or manipulation. It therefore provides a framework not only for assessing current exposure, but also for examining how the impact of AI on jobs may change over time under different technological trajectories.

A comprehensive forward-looking analysis could build on OECD's AI trajectory scenarios through 2030 (Hobbs et al., 2026[30]) that specify alternative patterns of capability progress across domains and over time, while recognising that such scenarios are illustrative rather than predictive. The four scenarios (Progress Stalls, Progress Slows, Progress Continues, and Progress Accelerates) detail possible different levels of future AI across the capability indicators. There are also variants of each of the scenarios that showcase more uneven progress across different indicators. The scenarios are not predictions, but illustrations of possible alternative futures. The 2030 Trajectory scenarios only look at cutting-edge capabilities and do not discuss the diffusion of technologies into society, which may be significantly delayed. These scenarios could support future research to estimate potential future occupational exposure under different assumptions about the rate of progress in different capability domains. In future work, such capability scenarios could also serve as inputs into macroeconomic models of AI-driven productivity and structural change, helping to connect occupational exposure analysis with wider economic projections.

The scenario presented below is an illustrative exercise in changing AI capabilities rather than a prediction of adoption or occupational outcomes. It is intended to show how the measure can be used to explore alternative technological paths, while recognising that actual workplace effects will also depend on diffusion and notably the cost of AI and robotics systems (as compared to the cost of human operators), organisational uptake, regulation, social choice and broader decisions about how technology is integrated into work.

Under a scenario in which AI's cognitive capabilities increase by one level, the largest reductions in capability gaps are concentrated in occupations that rely heavily on social interaction, problem solving, metacognition, and knowledge-related capabilities (Table 7.1. 7.1). As a result, the overall AI Capability Gap index declines further in a number of cognitive and knowledge-intensive occupational groups, while changing comparatively little in occupations where physical capabilities remain the main bottleneck.

Table 7.1. Gaps between occupational capability demands and future AI capabilities and overall AI Capability Gap index in a scenario of AI cognitive capabilities increasing by one level

Major Group	Healthcare Practitioners and Technical	Protective Service	Community and Social Service	Legal	Educational Instruction and Library	Management	Business and Financial Operations	Computer and Mathematical	Sales and Related	Life, Physical, and Social Science	Arts, Design, Entertainment, Sports, and	Architecture and Engineering	Personal Care and Service	Healthcare Support	Food Preparation and Serving Related	Installation, Maintenance, and Repair	Construction and Extraction	Farming, Fishing, and Forestry	Building and Grounds Cleaning and Maintenance	Transportation and Material Moving	Production	Office and Administrative Support	
Major Group Code	**29**	**33**	**21**	**23**	**25**	**11**	**13**	**15**	**41**	**19**	**27**	**17**	**39**	**31**	**35**	**49**	**47**	**45**	**37**	**53**	**51**	**43**	
Language	0.0	0.0	0.2	0.3	0.3	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**
Social interaction	0.7	0.7	1.3	0.8	1.0	0.8	0.5	0.1	0.6	0.4	0.6	0.1	0.6	0.4	0.3	0.0	0.0	0.1	0.1	0.1	0.0	0.1	**0.4**
Problem solving	0.6	0.8	1.2	0.9	0.9	1.0	0.5	0.4	0.2	0.8	0.4	0.4	0.3	0.1	0.2	0.0	0.0	0.1	0.1	0.2	0.0	0.0	**0.4**
Creativity	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**
Metacognition	0.6	0.6	0.9	1.0	1.0	1.0	0.5	0.5	0.2	0.7	0.5	0.5	0.1	0.0	0.1	0.0	0.0	0.1	0.0	0.1	0.0	0.0	**0.4**
Knowledge, learning and memory	0.2	0.0	0.1	0.5	0.4	0.1	0.0	0.1	0.0	0.3	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.1**
Vision	0.5	0.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.5	0.1	0.3	0.2	0.2	0.4	0.5	0.6	0.5	0.4	0.3	0.0	**0.2**
Manipulation	1.3	0.8	0.0	0.0	0.1	0.1	0.0	0.0	0.2	0.4	0.5	0.4	1.1	1.6	1.3	1.8	1.6	1.2	1.4	0.8	1.4	0.1	**0.7**
Robotic intelligence	0.9	1.4	0.0	0.0	0.2	0.0	0.0	0.0	0.2	0.3	0.3	0.2	1.1	1.3	1.1	1.3	1.5	1.3	1.3	1.1	0.5	0.1	**0.6**
Overall AI Capability Gap Index	**2.8**	**2.8**	**2.7**	**2.7**	**2.4**	**2.2**	**1.1**	**0.8**	**0.8**	**1.9**	**1.6**	**1.1**	**1.7**	**1.8**	**1.6**	**2.6**	**2.6**	**2.2**	**2.0**	**1.9**	**1.5**	**0.2**	

The results suggest that stronger cognitive AI would broaden exposure beyond clerical and routine office work. Office and Administrative Support Occupations reach a total capability gap index of 0.2, indicating that, under this scenario, current remaining cognitive gaps would be effectively closed for this occupational group. Computer and Mathematical, Sales and Related Occupations, Business and Financial Operations, and Architecture and Engineering also record relatively low indexes, all between 0.8 and 1.1. Management and Life, Physical, and Social Science Occupations fall to 2.2 and 1.9 respectively, indicating that progress in cognitive capabilities would bring AI substantially closer to the requirements of occupations that were previously much less exposed.

At the same time, the results show that a significant set of occupations would remain comparatively less exposed even in this scenario. The highest remaining gaps are found in Healthcare Practitioners and Technical and Protective Service Occupations, both at 2.8, followed by Community and Social Service and Legal Occupations (both 2.7) and by Construction and Extraction and Installation, Maintenance, and Repair Occupations (both 2.6). For Community and Social Service and Legal Occupations, the residual distance is almost entirely cognitive, reflecting persistent gaps in social interaction, problem solving and metacognition rather than any physical demands. For Construction and Extraction and Installation, Maintenance, and Repair, it is the opposite, lying almost wholly in manipulation and robotic intelligence. Healthcare Practitioners and Protective Service Occupations record the largest gaps of all precisely because they combine the two—substantial manipulation and robotic-intelligence requirements alongside continued reliance on judgement and social interaction—so that gains in cognitive capabilities alone leave a large part of their capability profile unaddressed.

The case study shows how the AI Capability Gap index can be used to trace changing patterns of occupational exposure under alternative AI development paths. In this scenario, advances in cognitive capabilities would extend AI exposure further into professional, technical and analytical work. However, they would not eliminate important constraints in occupations where performance depends on physical interaction with the environment, fine motor skills, or real-time adaptation in unpredictable settings. The results therefore underline that future AI impacts are likely to depend not only on how fast AI improves, but also on which capabilities improve first.

Next steps

The AI Capability Gap index presented here is intended as a first important measure towards a more flexible and policy-relevant framework for assessing the impact of AI on work. Several extensions would strengthen its analytical value.

A first priority is to move from occupations to tasks. Occupations bundle together a wide range of activities, and the capability demands within the same occupation can vary substantially across tasks. Therefore, identifying a gap in one capability domain, as with the current capability gap index, does not necessarily indicate whether AI can perform an occupational task effectively. A task-level AI Capability Gap index would therefore provide a more precise picture of where AI is likely to substitute for human labour, where it may complement workers, and where little change is expected. Such an approach would also make it possible to identify forms of within-occupation transformation that are not visible in occupation-level averages. This is particularly important for occupations in which a limited set of tasks is highly exposed, while other core tasks continue to depend on capabilities that remain beyond the AI frontier. A task-based extension would therefore allow a more granular analysis of job redesign, task reallocation and changing skill needs.

A second priority is to extend the analysis from occupational and task change to its implications for education and training systems. Once the tasks and capabilities most affected by AI are identified, the next step is to assess how this changing demand profile should be reflected in curricula, qualifications and lifelong learning provision. That kind of extension would need to be combined with explicit assumptions about future AI trajectories rather than relying only on current exposure estimates. This would make it possible to distinguish between capabilities that may decline in relative importance because AI can increasingly perform the associated tasks, and those that are likely to become more valuable because they remain complementary to AI or continue to lie beyond its frontier. Such an extension would help policymakers move from measuring exposure to anticipating how initial education, vocational education and training, higher education and adult learning systems may need to adapt. In this sense, linking the AI Capability Gap index to educational content would provide a second layer of policy relevance, supporting more forward-looking decisions on what people should learn, how learning pathways should evolve, and which human capabilities education systems should prioritise in a context of rapid AI progress.

A third step is to aggregate the AI Capability Gap index beyond occupations. Because the measure is defined at the occupational level, it can be combined with employment structures to construct indicators at the industry level. This would make it possible to compare the potential exposure of sectors depending on their occupational composition and to identify where AI may have the greatest effects on production processes, business models and labour demand. Industry-level aggregation would also support policy analysis by highlighting sectoral differences in adjustment needs, training requirements and the likely pace of organisational change. More broadly, such aggregation would extend the usefulness of the index from the study of jobs to the study of structural change across the economy. These analyses would build upon and extend previous sectoral-level OECD work (Calvino et al., 2024[31])

Another extension is to calculate country-specific and region-specific AI Capability Gap indexes. While the capability requirements of occupations may be broadly comparable across countries and regions, labour markets differ in occupational mixes, sectoral specialisation, workforce skills, digital infrastructure and firms' capacity to adopt AI. As a result, the implications of a given level of AI capability are unlikely to be uniform across countries and regions. Combining the occupational AI Capability Gap index with national and regional employment data would make it possible to estimate country-specific exposure profiles and to compare how the potential impact of AI varies across economies. This would provide a useful basis for cross-country and within-country analysis of labour market risks and opportunities, and for the design of national and regional strategies on skills, innovation and AI adoption.

One especially promising direction would be to combine the occupational exposure framework with the OECD's capability-based AI scenarios through 2030 (Hobbs et al., 2026[30])and with macroeconomic models of AI-driven productivity and structural change of the kind developed by (Filippucci, Gal and Schief, 2024[32]), (Filippucci, Gal and Schief, 2025[33]). This would make it possible to examine how different AI capability trajectories, combined with different assumptions about adoption and diffusion, may affect productivity, labour demand, sectoral composition and workforce adjustment, and could in time support integrated country-level analysis.

These extensions would turn the AI Capability Gap Index into a broader platform for analysing AI's labour market effects across different levels of observation. Task-level estimates would improve precision, industry-level aggregation would strengthen sectoral analysis, and country-specific measures would support international comparison. In this way, the framework could evolve from an occupational exposure indicator into a more general tool for scenario-based forward-looking analysis of how advances in AI may reshape work, production and labour markets. The measure should therefore be seen as a foundation for analysing potential occupational exposure, to be combined in future work with task-level evidence, diffusion assumptions, and broader economic and institutional analysis.

Conclusion

This paper develops a new OECD measure of occupational AI exposure based on the OECD AI Capability Indicators. By comparing occupational capability requirements with the current AI frontier across cognitive, social and physical domains, it provides a framework that is forward-looking, transparent, updateable and broadly linked to human abilities, rather than relying on a single proxy for AI progress or a one-dimensional view of exposure.

The results show that current AI exposure is highest in occupations with more standardised and codifiable task content, while substantial gaps remain in domains such as social interaction, problem solving, metacognition and many forms of embodied performance. They also show that AI exposure is not one-dimensional: occupations differ not only in how exposed they are, but in which capabilities are driving that exposure.

The measure should be interpreted as a capability-based indicator of potential exposure rather than as a direct measure of AI adoption or realised labour-market effects. It provides a useful summary of occupational capability gaps, but its composite index should be read alongside the underlying domain-specific results and in combination with evidence on diffusion and most notably the cost of AI and robotics systems, task structure and institutional context.

Taken together, the OECD AI Capability Gap Index offers a foundation for richer analysis of how AI may reshape work, skills and the economy. Future work can build on it through task-level extensions, scenario-based applications, and integration with broader macroeconomic and country-level analysis.

Annex 1. Sample selection and occupational sorting procedure

Given the scale of evaluating all 900 occupations in the O*NET database, a full manual assessment was beyond practical limits. To manage this challenge effectively, the mapping was done in two steps, initially mapping a sample of occupations by hand and then extending that mapping to the full set of occupations using LLMs. A strategically selected sample of occupations was used for the initial mapping done by hand to ensure that there were examples of human-rated occupations across the full diversity of capability profiles found in the workforce.

To identify an appropriately diverse sample of occupations for the initial human rating, the OECD collaborated with occupational psychologists led by Phillip Ackerman at Georgia Tech. Occupations were classified based on broad characteristics of the required knowledge, skills and abilities (KSAs). The focus was on four broad KSAs categories: social interaction, reasoning, physical abilities, and sensory/perceptual capabilities. O*NET was instrumental in identifying occupations with high KSAs in these areas. The selection process aimed for a diverse representation of occupations across the four KSA categories, as well as all knowledge domains, occupational groups and job zones (which capture different levels of training and experience).

In the first stage, 60 occupations were selected based on their relevance to the four KSA categories, and 8 additional occupations were included because of their high employment prevalence in the United States, as a broad indication of their prevalence in OECD economies generally. For each of the resulting 68 occupations, three to four key tasks were extracted from O*NET, prioritising those considered most important for the occupation and most representative of the targeted capability demands.

In a second stage, the major task for each occupation was evaluated by two groups of expert judges (industrial/organisational psychology and computer science graduate students), together with OECD analysts. Judges grouped tasks and correspondingly occupations according to perceived similarity, generating co-occurrence matrices that were aggregated and analysed using multidimensional scaling and hierarchical clustering to identify patterns of similarity. Judges also provided ratings on specific capability dimensions (physical, reasoning, and social interaction) to better understand the underlying perceptual structure of the occupational space.

Finally, combining the results of the sorting exercise, the ratings, and additional OECD analysis, a subset of 40 tasks representative of 40 occupations was selected. These occupations were chosen to maximise diversity in capability profiles, minimise redundancy, and maintain representativeness across occupational groups and job zones. As a result, the final sample captures the main capability dimensions present in the O*NET occupational space and provides a suitable basis for analysing job requirements using the AI Capability Indicators (Table A1.1).

Table A1.1. Categories of distinct capability profiles

Reasoning ability demands	Social interaction demands	Physical, incl. vision ability demands	Capability profile description	Example occupations
High	Medium	High	High reasoning and physical demands,	Anesthesiologists, Radiologists, Dentists

Reasoning ability demands	Social interaction demands	Physical, incl. vision ability demands	Capability profile description	Example occupations
			some level of social interaction demands	
High	High	Low	High reasoning and social demands	Teachers, Marriage and Family Therapists
High	Medium	Medium	High reasoning, medium social and physical demands	Police Identification and Records Officers, Surveyors, Allergologists, Nursing Assistants
High	Medium	Low	High reasoning and medium-high social demands	Accountants, Financial Examiners
High	Low	Medium	High reasoning demands, medium physical demands	Life, Physical and Social Science occupations
Medium	High	High	High social and physical demands, medium reasoning demands	Firefighters, Massage Therapists
Medium	High	Medium	High social demands, medium reasoning and physical demands	Personal care and services occupations
Medium	High	Low	High social demands, medium reasoning demands	Concierges, Retail salespersons, Telemarketers
Medium-High	Medium	Medium	Medium-high reasoning demands, medium social and physical demands	Designers, Childcare workers
Medium	Medium	Medium	Medium reasoning and physical demands, some social demands	Bus Drivers, Airline Pilots, Crane and Tower Operators
Medium	Low	Low	Medium reasoning demands	Cargo and Freight Agents, Data Entry Keyers
Medium	Low	High	High physical demands, medium reasoning and low social interaction demands	Roofers, Jewellers, Electricians, Automotive Repair Mechanics
Low	Low	Medium	Medium physical demands, low reasoning and social demands	Laundry and dry-cleaning workers, Derrick operators, Meat, poultry & fish cutters & trimmers, Combined food preparation and serving workers, including fast food

Each capability profile blends the demands for reasoning, social interaction and physical (including perceptual) capabilities into distinct combinations that reflect real-world job requirements. While not every theoretical combination of high, medium and low demands is represented, this omission is justifiable. Most notably, profiles with low reasoning demands across the board are scarce or absent. This aligns with the nature of modern occupational tasks that typically involve at least some degree of cognitive engagement. It is difficult to imagine many jobs that require significant social or physical activity but virtually no reasoning. Even so-called routine and manual roles often involve reasoning. This reflects the inherently cognitive

nature of human work, affirming that the profiles included in the table cover the spectrum of practically relevant and existing job types in a meaningful and realistic way.

Annex 2. Detailed manual rating procedure

The first stage of the analysis asked 15 analysts from five different OECD directorates to individually link occupational requirements with the AI Capability Indicators. The objective was to generate a structured dataset connecting current occupational demands to the capability dimensions used to describe AI systems. This mapping represents a first step toward analysing how existing and emerging AI capabilities may relate to the labour market and occupational structures.

The exercise consisted of rating the capability requirements of the selected 40 occupations with respect to the nine AI Capability Indicators, resulting in a total of 360 individual ratings per rater. Analysts were asked to evaluate, for each occupation, the level of capability required (from 0 to 5) for each indicator. A value of 0 indicated an analyst's judgment that a specific capability was not required for performing the occupation, while higher levels represented a judgment of increasing complexity of the capability needed. The ratings focused on essential capability requirements, meaning the minimal level of ability necessary to carry out the occupation effectively.

To support the evaluation process, raters were provided with a set of background materials describing each occupation. These materials included a sentence-long occupational description, the corresponding job zone, a selection of relevant O*NET abilities and skills ranked by importance, and a condensed list of an occupation's constituent tasks. These elements were compiled in a document containing one page per occupation, with occupations listed alphabetically. Raters were instructed to consider all available information when determining the appropriate capability level for each indicator.

In addition, raters relied on detailed level descriptions for each AI Capability Indicator (OECD, 2025[21]), which defined the progression of capability complexity from level 1 to level 5. For the purposes of the exercise, raters were asked to interpret these descriptions by conceptually replacing references to AI systems or robots with human workers, ensuring that the ratings reflected the human capability requirements associated with each occupation. The process was designed to be iterative, allowing raters to adjust earlier scores as they progressed through the occupations in order to maintain internal consistency across ratings.

Additional guidance was provided for specific indicators where needed. For example, some capabilities - such as vision or manipulation - are common human abilities but are not essential for many occupations; therefore, they could receive a rating of zero when they were not necessary for performing the job.

Each rater spent approximately 10-20 hours on the individual mapping exercise and filled in a table with 9 rows for each of the AI Capability Indicators and 40 columns for each of the occupations with discrete values ranging between 0 and 5. Since two groups of raters (one consisting of 3 and the other of 2 analysts) consolidated their individual ratings into a single table per team, the result was 12 filled-in individual tables instead of 15.

Once the individual rating process had been completed, the initial OECD mean ratings were calculated, after which the group of OECD analysts engaged in a Delphi-like process to reach a collective consensus. Table A2.1 presents these initial OECD mean ratings, with occupations ranked from those with the highest overall capability demands to those with the lowest. The consequent analysis will follow the same order of occupations.

Table A2.1. Initial OECD mean ratings

Average level	Anesthesiologists	Dentists, General	Firefighters	Police Identification and Records Offic	Elementary School Teachers, Except Sp	Nursing Assistants	Nannies	Childcare Workers	Massage Therapists	Jewelers and Precious Stone and Metal	Automotive Service Technicians and M	Electricians	Roofers	Derrick Operators, Oil and Gas	Excavating and Loading Machine and D	Crane and Tower Operators	Bus Drivers, Transit and Intercity	Meat, Poultry, and Fish Cutters and Trir	Janitors and Cleaners, Except Maids ar	Laundry and Dry Cleaning Workers	Cargo and Freight Agents	Fast Food and Counter Workers	Foresters	Surveyors	Chemists	Geographers	Airline Pilots, Copilots, and Flight Engin	Allergists and Immunologists	Radiologists	Fashion Designers	Video Game Designers	Tutors	Marriage and Family Therapists	Financial Examiners	Accountants and Auditors	Conciergers	Retail Salespersons	Telemarketers	Court Reporters and Simultaneous Cap	Gambling Cage Workers
Language	4.2	4.1	3.6	3.8	4.7	3.8	4.3	4.6	3.2	2.4	2.4	2.4	1.5	2.2	1.5	1.7	2.3	0.9	1.3	1.1	2.7	2	3	3	3.6	3.8	3.9	4.2	4	3.5	3.8	4.5	4.7	3.6	3.3	3.5	4	4.4	3.6	2.2
Social Interaction	3.8	3.5	3.8	3.3	4.8	3.9	4.5	4.8	2.9	1.3	1.8	2.1	1.1	1.8	1.8	1.4	2.2	0.5	0.8	0.6	1.7	2.4	2.2	1.9	2.2	2.3	3.3	3.7	3.1	3.7	3.3	4.2	4.8	2.1	1.7	3.8	4	3.7	1.5	2.2
Problem Solving	4.2	4.1	4.8	4.1	4.6	3.5	3.8	4.2	2.8	2.3	3.1	3.1	2.5	2.8	2.1	1.9	2.3	1	1.3	1.1	2	1.5	2.7	2.8	3.6	3.6	4.8	4.2	3.7	3.8	3.5	3.9	4.7	3.4	2.6	3	2.5	2.2	1.2	1.3
Creativity	2.3	2.1	2.5	2.2	2.8	1.8	2.2	2.7	1.4	3.4	1.8	1.7	1.3	0.9	1.2	0.8	0.9	0.4	0.5	0.4	1.2	0.5	2.1	1.4	3.1	3	2.1	2.4	2	4.8	4.8	2.6	3.3	1.8	1.4	1.8	1.5	1.6	0.6	0.5
Metacognition	3.9	3.5	4.4	3.6	3.5	3.2	3.3	3.5	2.3	2.4	2.1	2.2	1.9	2.1	1.4	1.5	1.9	1.1	1.3	0.9	1.6	1	2.9	2.8	3.3	3.8	4.4	3.8	3.8	3.2	3.1	3.3	4	3.2	2.8	2.2	1.9	1.9	1.8	1.3
Knowledge, Learning and Memory	4.6	4.2	3.8	4.1	3.7	3.3	2.9	3.1	2.7	2.9	2.8	3	1.8	2.5	1.8	1.5	2.4	1.3	1.5	1.5	2.3	1.3	3.3	3.1	4	4.2	4.3	4.4	4.3	3.4	3.4	4	3.7	3.8	3.3	2.2	2	2	1.8	1.8
Vision	3.2	4.3	4.8	4.2	2.9	4	4.1	4.5	1.8	3.9	3.8	3.9	3.5	3.8	3.9	4	4.6	2.8	3.1	2.7	2.7	2.2	3.8	3.8	2.9	3.4	4.8	2.9	4	3.8	3.1	0.4	1	0.3	0.2	2.2	2.2	0	0.8	1.5
Manipulation	3.8	4.8	4.8	2.9	3	4.5	4.3	4.7	4.3	4.3	4.3	4.5	4.3	4.4	3.8	3.5	1.8	3.4	3.3	2.7	1.9	2.6	2.6	2.3	3.2	1.3	2	2	1.6	2.3	0.3	0	0	0.1	0	1.7	1.8	0	0.5	1.8
Robotic Intelligence	3.9	4.1	4.9	3.5	3.8	4.4	4.8	4.7	3.2	2.2	3.6	3.9	4.1	3.9	4.1	3.3	4.2	2.5	3	2.1	2.5	2.5	3.3	3.5	3.1	1.5	3.4	2.6	1.3	1.8	0	0	0.2	0.2	0.3	1.7	2.3	0	0.3	1

Annex 3. Consensus process and convergence diagnostics

Initial methodological challenges to the rating process

The rating exercise proved to be a substantive and intellectually demanding task and the raters identified several recurring sources of difficulty that shaped both individual judgments and collective discussions. These challenges can be broadly grouped around the nature of occupations themselves, the structure of the AI capability indicators and the interpretive dynamics that emerged during the rating process.

Fluid Nature of Occupational Tasks

A fundamental tension in the exercise concerned the inherent instability of task bundles across occupations. The instructions asked raters to assess the level of AI capability required to perform a given occupation as it is performed today. However, this proved difficult in practice, because occupations are not fixed sets of tasks: they are shaped by technological change, societal expectations and the particular context in which work is performed. Several raters observed that tasks currently bundled within a single occupation may become unbundled as AI and automation progress, making it unclear whether the rating should reflect current task requirements or anticipate foreseeable shifts in what a role entails.

A frequently cited illustration involved bus drivers: today, the occupation encompasses driving, cleaning, and handling luggage, all performed by a human. As autonomous vehicles become viable, these tasks are likely to be disaggregated, leaving open questions about which capability requirements are meaningful for the purpose of the rating. This type of uncertainty was most visible in the physical capability indicators, particularly Manipulation and Robotic Intelligence, where raters often disagreed not about the current structure of the job, but about whether to anchor their rating to present-day task performance or to a plausible near-future configuration of the occupation.

Related to this was a concern about what might be called “desirable” versus “actual” performance: some raters tended to score occupations in line with how they believed a job ought to be performed, rather than how it is typically carried out. For instance, expectations about empathy and social capability for healthcare practitioners varied considerably depending on the normative lens through which raters approached the exercise.

Overlaps Between AI Capability Indicators

A second major challenge concerned the conceptual boundaries between the AI capability indicators themselves. Several raters found it difficult to assign distinct scores to indicators that, at certain levels, appeared to describe overlapping cognitive or physical competencies. The distinction between Manipulation and Robotic Intelligence, for example, was not always clear: some raters interpreted Robotic Intelligence as a holistic integration of all physical capabilities, while others treated it as a more narrowly defined physical navigation capacity. These divergent interpretations produced systematically different scoring patterns across the group.

Similarly, Critical Thinking appeared both within the Language scale, where higher levels reference nuanced reasoning and argumentation, and within the Metacognition and Critical Thinking indicator,

creating a risk of overemphasising this capability for occupations that require complex discourse. The level 5 descriptor for Robotic Intelligence compounded this by referencing common sense, social context, and general knowledge, qualities that nominally also belong to other indicators such as Social Interaction and Problem Solving. This structural ambiguity in the scales made it genuinely difficult for raters to isolate what each indicator was measuring, particularly at the upper levels, and introduced a risk of double-counting capability demands across indicators.

Methodological Tensions in Scale Interpretation

Beyond the specific challenges posed by individual occupations or indicators, raters confronted a more general tension about how to use the scales at all. Two competing interpretive logics emerged. The first involved strict adherence to the textual description of each level, using it as the primary reference for assigning a score. The advantage of this approach is that it maximises fidelity to the scale as designed, but in some cases raters had a hard time understanding how to match the scale to the demands of the occupation. The second logic involved incremental or relational scoring. For example, if one occupation received a score of 4 on Language, a clearly more demanding occupation would receive a 5. This approach produces greater internal consistency and comparability across the dataset, but risks losing anchoring to the actual meaning of each level.

Most raters navigated this tension pragmatically, defaulting to the textual descriptions where these were informative and falling back on relative comparisons where they were not. However, the lack of a shared and explicit protocol for handling this trade-off introduced variability. The richness of scale descriptions also varied substantially: some indicators provided detailed level descriptions with illustrative task examples, while others offered only brief or abstract characterisations, particularly at the higher levels. This unevenness meant that the mapping exercise was more tractable for some indicators than others, independent of the actual complexity of the occupations being rated.

A further methodological difficulty arose from the uneven spacing between levels within a given scale. Raters noted, for instance, that the distance between levels 3 and 4 on the Language scale felt substantially larger than the equivalent step on the Metacognition scale. This inconsistency across scales complicates the aggregation of scores into composite indices, since a one-unit difference does not carry the same substantive meaning across all indicators.

Finally, raters observed a recalibration effect during the exercise itself. Having rated dozens of occupations over an extended period, many found that their interpretive reference points shifted, prompting them to revise earlier scores for consistency after completing a larger portion of the dataset. While averaging across raters partially addresses this within individual scores, it underscores the importance of collective calibration early in the process.

Methodological clarifications during the discussion

Following the initial round of individual ratings, the group convened to discuss cases of significant disagreement and to develop shared interpretive norms. This process unfolded across multiple meetings and involved two complementary forms of clarification: deepening understanding of specific occupations, and refining the interpretation of particular capability indicators.

Improving the Understanding of Some Occupations

One productive approach to reducing disagreement was to examine particular occupations in detail and trace the source of divergence in ratings. In many cases, high dispersion reflected not a genuine difference

in how raters understood AI capabilities, but rather different assumptions about which tasks were central to an occupation. Several cases illustrate this well.

For occupations involving physical tasks, such as foresters, bus drivers, or radiologists, raters differed substantially on Manipulation and Robotic Intelligence scores depending on whether they focused on the most physically demanding tasks listed in the occupational description or on tasks they deemed to be more representative of typical day-to-day work. Discussion allowed the group to agree on a shared understanding of which tasks should anchor the rating, typically converging on a level that reflected the modal task profile rather than the extreme cases.

In other instances, dispersion arose from oversights: a rater might have overlooked a key task in the occupational description, such as a buying and negotiating role in a jewellery occupation that substantially changed the appropriate Social Interaction score. In such cases, collective review allowed errors to be corrected without requiring a fundamental reinterpretation of the scale.

A more persistent form of disagreement involved occupations where the relevant tasks are genuinely contested or evolving. Bus drivers, airline pilots, and similar occupations prompted discussions about whether the exercise was intended to capture what a human currently does, or what an AI system would need to do to replace that human under any plausible future arrangement. No fully satisfactory resolution was reached for these cases, though the group generally agreed to prioritise ratings anchored in current task performance, leaving questions about technological displacement as a matter for subsequent analytical interpretation.

To support this process of collective calibration, several participants suggested ordering occupations by their average score on each indicator and inspecting the resulting ranking for inconsistencies. Where an occupation appeared to be clearly out of place relative to its neighbours in the ranked list, this served as a useful signal for further discussion. This approach was seen as particularly valuable not only for the initial set of rated occupations, but also as a quality check on the expanded dataset generated for the full occupational classification.

Improving the Understanding of Some AI Capability Indicators

In parallel with the discussion of individual occupations, the group invested considerable effort in clarifying how specific indicators should be interpreted. Several indicators proved to be persistent sources of confusion and were discussed at length across multiple sessions.

The Language indicator posed a specific challenge around the “dialogue” example associated with level 4. Because dialogue appears as an illustrative task at that level, raters who interpreted it literally felt compelled to assign a 4 to any occupation that involved any form of active conversation, including occupations where dialogue represents only a small and peripheral part of the role. The group discussed the importance of interpreting such examples contextually rather than mechanically, reserving level 4 for occupations where active, substantive dialogue is a defining feature of the work rather than an occasional task.

The Creativity indicator generated significant debate, particularly regarding the appropriate ceiling for occupations in health and education. One rater consistently assigned high creativity levels to these roles, reasoning that professional judgment and academic research always involves some degree of creative adaptation. Others in the group argued that creativity at the higher levels should be reserved for occupations where generating genuinely novel outputs is a core requirement, not for occupations where the expectation is that practitioners follow established, evidence-based approaches, adapting them only marginally to unusual circumstances. Following discussion, the group agreed to recalibrate creativity ratings across occupations and to ensure that the upper levels of the scale were not used too liberally.

The Knowledge, Learning, and Memory (KLM) indicator was another area of sustained disagreement. One rater argued that any occupation involving real-time interaction with humans in natural environments requires episodic memory and continuous learning, and assigned correspondingly high scores to a wide range of occupations. The rest of the group tended to score more conservatively. This divergence prompted a careful collective re-reading of the level descriptions and a discussion of what should count as meaningful differences across levels, leading to a degree of harmonisation across occupations.

The Robotic Intelligence indicator remained among the most difficult to interpret consistently throughout the process. Raters adopted quite different frameworks: some treated it as a composite indicator capturing the overall physical intelligence needed to perform the job, integrating vision, manipulation, and environmental navigation; others interpreted it more narrowly as a measure of autonomous physical movement. The absence of a shared resolution meant that scores on this indicator tended to show the highest dispersion across the group, and several cases were flagged for individual post-hoc review rather than resolved through collective agreement.

Across all these discussions, a recurring theme was the need to distinguish between two types of disagreement. The first type reflects genuine substantive uncertainty, cases where multiple interpretations are plausible and where the diversity of views may itself be informative about the contested nature of an occupation's AI capability profile. The second type reflects interpretive misalignment, cases where raters applied the scales in ways that were inconsistent with the shared instructions, or where they overlooked relevant occupational information. The group generally agreed that the first type of disagreement should be preserved and reflected in the uncertainty of the final ratings, while the second type should be identified and corrected as far as possible. Measures of dispersion such as standard deviation and maximum score differences were used to flag cases where further scrutiny was warranted.

These discussions were useful not only for improving consistency in the rating exercise, but also for informing the forthcoming revision of the OECD AI Capability Indicators. They highlighted a number of inconsistencies, overlaps and ambiguities in the current descriptions that the revision aims to address through clearer wording and sharper conceptual boundaries. If some scales are significantly altered, a partial revision of the estimates of occupational demands might be needed.

Evolution of consensus finding

The evolution of the ratings across rounds points to a clear two-stage consensus process. The first stage, which followed the initial individual ratings and the first collective discussions, involved substantial recalibration of both the level of ratings and the degree of agreement among raters. The second stage, after a renewed discussion, was much more limited in scope and mainly served to consolidate a set of assessments that had already largely converged. This pattern is consistent with the broader Delphi-like logic of the exercise, in which discussion is used first to align interpretations of occupations and indicator scales, and only later to resolve the remaining outliers and ambiguities.

The strongest evidence of convergence comes from the evolution of rating dispersion. Across the 360 occupation-indicator cells, the average standard deviation fell from 0.98 in the initial round to 0.80 after the first revision, and then only marginally further to 0.78 in the final round (Table A3.1). The number of cells with a standard deviation above 1 also declined sharply, from 139 initially to 78 after the first revision and 53 in the final ratings. This indicates that the main gain in consensus occurred early in the process. The later stage still produced some additional tightening, but the scale of change was smaller.

Table A3.1. Evolution of Standard Deviation of OECD ratings

Standard Deviation of initial ratings	Anesthesiologists	Dentists, General	Firefighters	Police Identification and Records Officer	Elementary School Teachers, Except Spe	Nursing Assistants	Nannies	Childcare Workers	Massage Therapists	Jewelers and Precious Stone and Metal W	Automotive Service Technicians and Mec	Electricians	Roofers	Derrick Operators, Oil and Gas	Excavating and Loading Machine and Dra	Crane and Tower Operators	Bus Drivers, Transit and Intercity	Meat, Poultry, and Fish Cutters and Trimm	Janitors and Cleaners, Except Maids and	Laundry and Dry Cleaning Workers	Cargo and Freight Agents	Fast Food and Counter Workers	Foresters	Surveyors	Chemists	Geographers	Airline Pilots, Copilots, and Flight Enginee	Allergists and Immunologists	Radiologists	Fashion Designers	Video Game Designers	Tutors	Marriage and Family Therapists	Financial Examiners	Accountants and Auditors	Concierges	Retail Salespersons	Telemarketers	Court Reporters and Simultaneous Captio	Gambling Cage Workers	**Average**
Language	1.0	0.9	1.1	1.1	0.7	1.0	0.9	0.5	1.3	0.8	0.7	0.7	1.2	1.0	1.0	0.9	0.5	1.0	1.2	0.9	0.5	0.6	1.1	0.6	0.8	0.6	0.8	0.7	0.7	0.8	0.9	0.7	0.5	1.0	0.7	0.5	0.9	0.7	1.4	0.7	**0.8**
Social Interaction	1.2	1.1	0.8	1.2	0.6	1.2	0.7	0.4	1.2	1.2	0.9	0.9	1.1	1.1	1.1	1.2	0.9	1.0	1.0	1.0	0.8	0.5	0.6	1.1	0.9	0.7	0.6	1.1	0.9	0.8	1.0	0.7	0.6	1.3	1.2	0.9	0.8	0.9	1.4	0.6	**0.9**
Problem Solving	0.9	0.7	0.5	1.1	0.5	1.3	1.5	1.5	1.0	0.8	1.2	1.0	0.8	1.2	1.2	1.1	1.2	0.7	1.1	0.9	1.0	1.0	1.2	1.1	0.9	1.1	0.6	0.7	1.0	0.8	1.5	0.8	0.5	1.0	1.4	1.2	0.9	0.9	1.1	0.8	**1.0**
Creativity	1.5	1.1	1.5	1.3	1.2	1.2	0.7	1.2	0.8	1.3	1.2	0.8	0.7	1.0	0.7	0.8	1.1	0.5	0.5	0.5	0.8	0.5	1.0	0.8	1.3	1.0	0.8	1.2	1.2	0.4	0.4	1.0	0.9	1.5	0.5	1.1	1.2	1.2	0.7	0.5	**0.9**
Metacognition	0.9	1.2	0.7	1.0	1.2	1.1	1.1	1.4	0.8	1.2	0.8	0.7	0.7	1.2	0.5	0.7	0.8	0.5	0.9	0.7	0.5	0.4	1.0	0.7	1.3	0.7	0.7	0.9	0.9	1.0	1.1	0.8	0.6	1.0	0.7	1.0	0.7	0.5	0.7	0.5	**0.8**
Knowledge, Learning and Memory	0.5	0.7	0.6	1.1	1.0	0.9	1.0	1.4	1.0	0.8	0.9	1.1	1.1	1.1	1.1	1.2	1.2	1.0	1.2	0.9	1.0	1.1	1.0	1.1	1.4	0.9	0.9	0.7	0.7	0.9	1.0	0.7	1.2	1.0	1.2	1.1	1.0	1.0	0.8	1.1	**1.0**
Vision	1.8	0.7	0.6	1.0	1.8	0.9	0.9	0.8	1.7	1.0	0.6	0.8	0.7	0.8	0.7	0.4	0.8	1.2	1.3	1.0	1.2	1.3	1.1	1.1	1.7	0.8	0.4	1.5	1.0	1.3	1.7	1.2	1.9	0.7	0.4	1.7	1.5	0.0	1.2	1.4	**1.1**
Manipulation	1.9	0.4	0.6	2.1	2.0	0.9	1.0	0.9	1.4	0.7	0.6	0.8	0.9	0.7	0.9	1.4	1.8	0.8	1.0	1.0	1.6	0.9	2.0	1.9	1.6	1.8	2.2	2.1	1.8	1.9	1.2	0.0	0.0	0.3	0.0	1.8	1.5	0.0	1.4	1.1	**1.2**
Robotic Intelligence	1.7	1.2	0.3	1.4	1.9	0.9	0.6	0.8	1.5	1.3	1.0	0.8	0.7	0.9	0.7	1.4	1.4	1.4	1.0	1.0	1.3	1.0	1.4	0.9	1.2	1.7	2.2	1.7	1.4	1.9	0.0	0.0	0.6	0.4	0.5	1.8	1.4	0.0	0.7	1.0	**1.1**
Average SD	**1.3**	**0.9**	**0.7**	**1.3**	**1.2**	**1.0**	**0.9**	**1.0**	**1.2**	**1.0**	**0.9**	**0.8**	**0.9**	**1.0**	**0.9**	**1.0**	**1.1**	**0.9**	**1.0**	**0.9**	**1.0**	**0.8**	**1.2**	**1.0**	**1.2**	**1.0**	**1.0**	**1.2**	**1.1**	**1.1**	**1.0**	**0.7**	**0.8**	**0.9**	**0.7**	**1.2**	**1.1**	**0.6**	**1.0**	**0.9**	**1.0**

Standard Deviation of first revision																																								
Language	0.7	0.5	1.1	1.1	0.7	1.0	0.9	0.5	0.8	0.9	0.7	0.7	1.0	0.9	0.8	0.9	0.4	0.9	1.2	0.9	0.5	0.7	0.9	0.6	0.7	0.6	0.5	0.6	0.4	0.8	0.8	0.5	0.5	1.0	0.7	0.5	0.6	0.8	1.3	0.7	**0.8**
Social Interaction	0.8	0.8	0.5	1.0	0.7	1.3	0.7	0.5	1.1	1.0	0.8	0.8	1.1	1.1	1.0	1.1	0.8	0.9	1.1	0.9	0.8	0.6	0.7	1.2	0.9	0.8	0.7	0.8	0.7	0.6	0.7	0.8	0.4	1.0	0.9	0.8	0.7	0.8	1.1	0.6	**0.8**
Problem Solving	0.6	0.5	0.4	0.9	0.7	1.2	0.8	0.8	0.8	0.8	0.7	0.9	0.8	0.9	0.8	0.8	0.8	0.7	1.0	0.9	0.6	0.9	1.0	0.9	0.9	1.0	0.6	0.5	0.5	0.7	0.9	0.8	0.5	0.5	0.6	0.5	0.5	0.9	0.7	0.7	**0.8**
Creativity	1.1	0.8	1.3	1.3	0.9	0.8	0.8	0.8	0.8	0.8	1.0	1.0	0.8	1.0	0.7	0.8	0.6	0.5	0.5	0.5	0.7	0.5	0.9	0.8	1.0	1.0	0.5	1.3	0.9	0.4	0.4	0.9	0.9	1.1	0.5	0.9	0.9	1.2	0.5	0.5	**0.8**
Metacognition	0.8	0.9	0.5	0.8	0.8	1.1	0.9	0.8	0.8	0.7	0.6	0.7	0.6	0.6	0.5	0.7	0.8	0.5	0.8	0.6	0.5	0.5	0.7	0.7	0.7	0.8	0.7	0.8	0.8	0.9	0.9	0.6	0.8	0.8	0.5	0.5	0.6	0.4	0.6	0.5	**0.7**
Knowledge, Learning and Memory	0.5	0.7	0.6	0.8	0.8	1.0	1.0	0.7	0.9	0.7	0.8	0.9	1.1	1.0	1.1	1.1	1.1	0.8	1.2	0.8	0.9	1.2	0.7	0.9	0.6	0.8	0.8	0.7	0.7	1.0	1.0	0.7	1.1	0.8	1.0	0.9	0.8	0.9	0.8	1.1	**0.9**
Vision	1.1	0.6	0.3	0.7	1.3	0.7	0.9	0.8	1.4	0.8	0.5	0.8	0.5	0.4	0.5	0.4	0.8	1.0	0.8	0.9	1.1	1.2	0.8	0.6	1.3	0.8	0.3	0.5	0.7	0.5	0.8	0.3	0.5	0.4	0.3	1.2	1.4	0.3	1.3	0.8	**0.8**
Manipulation	1.2	0.4	0.3	0.7	1.2	0.5	0.7	0.3	0.5	0.5	0.6	0.7	0.5	0.5	0.8	0.7	1.2	0.8	0.9	0.8	0.7	0.8	1.6	1.2	1.3	1.6	0.9	1.3	1.6	1.0	1.2	0.0	0.0	0.0	0.0	1.5	1.3	0.3	1.6	0.8	**0.8**
Robotic Intelligence	1.0	1.2	0.0	0.8	0.8	0.8	0.4	0.8	1.6	1.0	0.9	0.8	0.8	0.8	0.6	0.6	0.5	1.3	0.6	0.8	1.0	0.9	0.6	0.8	1.0	1.4	1.0	1.0	1.5	1.4	0.0	0.0	0.6	0.3	0.4	1.4	1.2	0.0	0.7	0.9	**0.8**
Average SD	**0.9**	**0.7**	**0.6**	**0.9**	**0.9**	**0.9**	**0.8**	**0.7**	**1.0**	**0.8**	**0.7**	**0.8**	**0.8**	**0.8**	**0.8**	**0.8**	**0.8**	**0.8**	**0.8**	**0.9**	**0.8**	**0.8**	**0.9**	**0.9**	**0.9**	**1.0**	**0.7**	**0.8**	**0.9**	**0.8**	**0.8**	**0.5**	**0.6**	**0.7**	**0.5**	**0.9**	**0.9**	**0.6**	**0.9**	**0.7**	**0.8**

Standard deviation of final ratings																																								
Language	0.7	0.6	1.1	1.1	0.7	1.0	0.9	0.5	0.9	0.9	0.7	0.7	1.0	1.0	0.8	0.9	0.4	0.9	1.0	0.9	0.5	0.7	0.8	0.6	0.7	0.6	0.5	0.6	0.4	0.8	0.8	0.5	0.5	1.0	0.7	0.5	0.7	0.7	1.2	0.7	**0.7**
Social Interaction	0.8	0.8	0.5	1.0	0.7	1.2	0.7	0.5	1.1	1.1	0.8	0.7	1.0	1.1	1.0	1.1	0.8	0.9	1.1	0.9	0.8	0.6	0.7	1.2	0.9	0.8	0.7	0.8	0.9	0.6	0.8	0.7	0.4	1.1	0.9	0.8	0.7	0.9	0.7	0.6	**0.8**
Problem Solving	0.6	0.6	0.5	0.9	0.7	1.2	0.8	0.8	0.8	0.8	0.9	1.0	0.8	0.9	0.8	0.7	0.8	0.7	1.0	0.9	0.7	0.9	1.0	0.9	0.9	1.0	0.6	0.5	0.5	0.8	0.9	0.8	0.5	0.7	0.7	0.6	0.5	0.9	0.7	0.7	**0.8**
Creativity	1.3	0.5	0.9	0.9	1.0	0.7	0.6	0.8	0.6	0.9	0.7	0.7	0.8	0.8	0.6	0.8	0.9	0.5	0.5	0.5	0.7	0.5	0.8	0.8	0.7	1.0	0.4	1.0	1.1	0.5	0.5	0.9	0.8	0.8	0.5	0.8	0.9	0.9	0.7	0.5	**0.7**
Metacognition	0.8	1.0	0.9	0.9	0.9	1.0	0.9	0.9	0.7	0.7	0.7	0.7	0.7	0.7	0.5	0.7	0.9	0.5	0.9	0.7	0.5	0.5	0.7	0.7	0.7	0.7	0.7	0.8	0.8	1.0	0.9	0.6	0.8	1.0	0.5	0.6	0.7	0.4	0.7	0.5	**0.7**
Knowledge, Learning and Memory	0.5	0.7	0.6	0.9	0.9	0.7	0.8	0.7	0.9	0.7	0.6	0.9	0.9	0.9	1.0	1.0	1.1	0.6	1.1	0.7	0.9	1.1	0.7	0.9	0.7	0.8	0.8	0.7	0.7	1.0	1.0	0.6	1.2	0.8	0.9	0.8	0.8	0.9	0.8	0.9	**0.8**
Vision	1.1	0.7	0.6	0.7	1.3	0.7	0.9	0.8	1.3	0.7	0.5	0.7	0.7	0.4	0.5	0.4	0.8	1.0	0.9	1.0	1.1	1.2	0.8	0.6	1.3	0.8	0.3	0.5	0.7	0.5	0.9	0.3	0.5	0.5	0.4	0.9	1.4	0.3	1.2	0.8	**0.8**
Manipulation	1.2	0.4	0.6	1.1	1.1	0.7	1.0	0.3	0.5	0.5	0.6	0.7	0.7	0.5	0.7	0.7	1.4	0.8	0.9	0.9	0.7	0.8	1.1	1.2	1.3	1.6	1.0	1.3	1.3	0.9	1.7	0.3	0.0	0.4	0.3	0.7	1.4	0.3	1.4	0.8	**0.8**
Robotic Intelligence	1.1	1.2	0.3	1.2	0.8	0.9	0.7	0.8	1.5	1.0	0.9	0.7	0.7	0.8	0.6	0.6	0.5	1.1	0.6	0.8	0.5	0.9	0.7	0.9	1.0	1.4	1.0	1.0	1.3	1.4	0.0	0.0	0.6	0.3	0.4	1.1	1.1	0.0	0.7	0.8	**0.8**
Average_SD	**0.9**	**0.7**	**0.7**	**1.0**	**0.9**	**0.9**	**0.8**	**0.7**	**0.9**	**0.8**	**0.7**	**0.7**	**0.8**	**0.8**	**0.7**	**0.8**	**0.8**	**0.8**	**0.9**	**0.8**	**0.7**	**0.8**	**0.8**	**0.9**	**0.9**	**1.0**	**0.7**	**0.8**	**0.9**	**0.8**	**0.8**	**0.5**	**0.6**	**0.7**	**0.6**	**0.8**	**0.9**	**0.6**	**0.9**	**0.7**	**0.8**

The first revision reduced disagreement especially strongly in the indicators that had already proved conceptually difficult. Average standard deviations fell most in Manipulation, Vision and Robotic Intelligence, the three indicators most directly linked to embodied and physical task interpretation. This is fully in line with the methodological issues identified above, where disagreement often arose from different views about which tasks should anchor the rating and from uncertainty about the conceptual boundaries between physical indicators.

The mean ratings show the same broad pattern (Table A3.2). Between the initial means and the first revision means, 17 occupation-indicator combinations changed by at least 0.5 capability levels, and 74 changed by at least 0.25. Between the first revision means and the final means, the corresponding numbers dropped to 3 and 12. The first round of revision therefore did not merely reduce noise around a stable centre: it shifted a meaningful number of collective judgements. By contrast, the final stage was mostly a fine-tuning exercise.

Table A3.2. Variation in OECD mean ratings

Difference between initial mean ratings and first revision	Anesthesiologists	Dentists, General	Firefighters	Police Identification and Records Officers	Elementary School Teachers, Except Special Edu	Nursing Assistants	Nannies	Childcare Workers	Massage Therapists	Jewelers and Precious Stone and Metal Workers	Automotive Service Technicians and Mechanics	Electricians	Roofers	Derrick Operators, Oil and Gas	Excavating and Loading Machine and Dragline Op	Crane and Tower Operators	Bus Drivers, Transit and Intercity	Meat, Poultry, and Fish Cutters and Trimmers	Janitors and Cleaners, Except Maids and Housek	Laundry and Dry Cleaning Workers	Cargo and Freight Agents	Fast Food and Counter Workers	Foresters	Surveyors	Chemists	Geographers	Airline Pilots, Copilots, and Flight Engineers	Allergists and Immunologists	Radiologists	Fashion Designers	Video Game Designers	Tutors	Marriage and Family Therapists	Financial Examiners	Accountants and Auditors	Concierges	Retail Salespersons	Telemarketers	Court Reporters and Simultaneous Captioners	Gambling Cage Workers
Language	-0.1	0.0	0.1	0.2	0.0	0.1	-0.2	0.1	0.3	-0.1	0.1	-0.1	-0.2	-0.3	0.0	0.1	-0.1	-0.3	0.0	0.0	0.0	0.1	0.2	0.3	0.0	0.2	0.2	0.0	0.2	0.1	-0.1	0.0	0.1	0.1	0.0	0.0	0.2	-0.2	-0.2	0.2
Social Interaction	-0.3	-0.3	-0.2	0.6	-0.1	-0.2	0.0	-0.1	-0.2	0.5	0.0	-0.1	0.1	-0.1	-0.1	0.0	0.1	0.0	0.0	0.0	0.2	-0.1	0.5	0.0	-0.1	0.0	0.1	-0.4	0.0	0.1	0.1	0.0	0.0	0.6	0.3	-0.1	-0.1	0.4	0.3	0.0
Problem Solving	0.0	0.0	0.1	0.2	-0.2	0.0	0.2	0.2	-0.2	0.0	0.0	0.1	0.1	0.2	0.6	0.3	0.2	-0.1	0.0	0.0	0.3	0.0	0.4	0.2	0.1	0.1	0.0	-0.3	0.2	-0.1	0.4	-0.1	0.0	0.1	0.1	0.1	0.2	0.2	-0.1	0.1
Creativity	0.1	-0.2	0.0	-0.1	-0.3	0.0	0.1	-0.3	0.0	0.5	-0.2	0.0	-0.1	-0.2	-0.1	-0.1	-0.2	-0.1	0.0	-0.1	-0.3	-0.2	0.1	0.0	0.1	0.1	-0.2	-0.1	-0.1	0.0	0.0	-0.2	-0.1	-0.3	-0.1	-0.2	-0.3	0.1	-0.3	0.0
Metacognition	-0.1	0.0	0.1	0.2	-0.1	0.0	0.0	0.0	-0.1	0.0	-0.1	0.2	0.1	-0.4	0.0	0.1	0.0	-0.2	0.1	0.1	0.1	0.1	0.0	0.1	0.3	0.0	-0.1	0.0	0.1	0.0	0.0	-0.1	-0.1	-0.1	-0.2	-0.1	0.1	0.1	0.0	0.0
Knowledge, Learning and Memory	-0.1	-0.1	0.0	0.3	-0.3	-0.1	-0.2	0.6	-0.1	0.0	0.0	-0.1	0.1	0.0	0.0	0.0	0.0	-0.1	-0.1	0.0	-0.1	0.0	0.0	0.3	0.0	0.3	0.3	0.1	-0.1	0.1	-0.1	-0.2	-0.1	0.1	0.1	-0.1	-0.1	0.3	0.1	0.1	0.1	0.1
Vision	0.5	-0.1	0.2	0.3	-0.2	-0.1	0.1	0.0	0.2	0.3	-0.1	0.1	0.1	0.1	0.0	0.0	0.1	0.2	0.4	0.1	0.0	0.2	0.1	0.0	-0.1	0.0	0.1	0.4	0.1	0.3	0.6	-0.3	-0.7	-0.1	-0.1	0.6	0.3	0.1	0.4	0.4
Manipulation	0.4	0.0	0.2	1.0	0.0	0.1	0.2	0.2	0.4	0.1	-0.1	0.1	0.3	0.0	0.2	0.4	0.4	0.1	0.0	0.1	0.0	0.2	0.4	0.8	0.1	0.8	0.9	1.0	0.7	0.9	0.0	0.0	0.0	-0.1	0.0	0.1	0.4	0.1	0.2	0.1
Robotic Intelligence	0.2	-0.2	0.1	0.4	0.2	-0.1	0.1	-0.1	0.1	0.1	-0.1	0.3	0.1	-0.2	-0.1	0.5	0.4	0.0	-0.2	0.3	0.2	0.1	0.5	0.1	-0.2	1.0	0.7	0.4	0.4	0.6	0.0	0.0	0.0	-0.1	-0.1	0.0	-0.1	0.0	0.0	0.2

Difference between first revision and final mean ratings																																								
Language	0.0	0.1	-0.1	-0.1	0.0	-0.1	0.0	-0.1	-0.1	0.0	0.0	0.1	0.1	0.1	0.0	-0.1	0.0	0.0	0.0	0.1	-0.1	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-0.1	-0.1	0.0	-0.1	0.0	0.0	0.0	-0.1	0.0	0.1	0.1
Social Interaction	0.0	0.1	0.0	-0.1	0.0	0.0	0.0	0.0	0.0	-0.2	0.1	0.0	0.0	0.0	-0.1	0.0	0.1	0.0	0.1	0.0	0.0	0.0	-0.1	0.0	0.0	0.0	0.0	0.0	0.1	-0.2	0.0	-0.1	0.0	0.0	0.1	-0.1	0.1	0.0	-0.1	-0.2	0.0
Problem Solving	0.1	0.1	-0.1	-0.1	0.1	0.0	-0.1	0.1	0.0	0.0	0.2	0.2	-0.1	-0.1	-0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	-0.1	0.0	-0.1	0.0	0.0	0.0	0.0	0.1	0.0	-0.1	-0.1	0.1	-0.1	0.1	-0.1	0.0	0.1	0.0
Creativity	0.2	-0.2	-0.3	-0.3	0.1	-0.1	0.0	0.0	-0.2	-0.2	-0.1	-0.1	0.0	-0.1	-0.1	0.0	0.2	0.0	0.0	0.0	-0.1	0.1	0.0	0.0	-0.2	0.0	-0.1	0.0	0.5	-0.3	-0.3	0.1	0.0	-0.1	0.0	0.0	0.1	-0.1	0.1	0.0
Metacognition	0.1	0.1	-0.2	-0.2	0.1	0.0	0.0	0.1	0.0	0.0	0.1	0.1	-0.1	0.1	0.0	0.0	0.1	0.0	-0.1	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.2	0.1	0.1	0.0	0.2	0.0	0.1	-0.1	0.0	0.1	0.0
Knowledge, Learning and Memory	0.0	0.1	-0.1	-0.1	0.1	-0.2	0.1	0.0	0.0	0.1	-0.1	0.0	0.1	0.0	0.1	0.2	0.1	0.1	0.1	0.1	0.1	0.0	0.1	0.1	0.1	0.0	0.1	0.1	0.0	0.1	0.1	-0.1	0.1	0.0	0.0	0.0	0.0	0.1	0.2	0.1
Vision	0.0	0.1	-0.2	0.0	0.1	-0.1	-0.1	0.0	-0.2	0.0	0.0	0.0	-0.1	0.0	0.0	0.0	0.0	-0.1	-0.1	-0.2	0.0	0.0	-0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.1	-0.1	0.0	0.0	0.1	0.1	-0.1	-0.1	0.0	0.0	-0.1
Manipulation	0.1	0.0	-0.2	-0.2	0.0	-0.1	-0.2	0.0	-0.1	0.0	0.1	0.0	-0.1	0.0	0.0	0.0	0.2	0.0	-0.1	-0.1	0.0	-0.1	-0.3	-0.1	0.1	-0.1	0.1	0.2	0.2	0.0	1.5	0.1	0.0	0.2	0.1	0.3	-0.2	0.0	0.4	-0.1
RoboticIntelligence	-0.2	-0.1	-0.1	-0.2	-0.1	-0.1	-0.2	0.0	-0.4	0.0	-0.1	0.0	0.0	0.0	0.0	-0.1	0.0	0.0	-0.1	0.0	-0.1	-0.1	-0.1	-0.1	0.0	0.1	0.1	-0.3	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.0

These first-round adjustments were concentrated in a small number of indicators. The largest average absolute shifts in mean ratings occurred in Manipulation, followed by Robotic Intelligence and Vision. Social Interaction and Problem Solving also moved appreciably, while Metacognition changed relatively little on average. This distribution is substantively plausible. It suggests that the early discussions were especially important for clarifying how to treat occupations with mixed physical and cognitive task bundles, and for deciding whether ratings should be anchored in the most demanding listed task or in the more typical profile of the occupation. That tension is described explicitly in the background material, particularly for occupations involving physical work or evolving task configurations.

At the occupational level, the strongest first-round recalibrations were concentrated in a limited subset of jobs. The highest average absolute shifts across indicators occurred for Police Identification and Records Officers, Allergists and Immunologists, Geographers, Foresters, and Airline Pilots, Copilots, and Flight Engineers. Several of the largest single-cell changes were especially striking. Manipulation increased by about one full level for both Allergists and Immunologists and Police Identification and Records Officers; Robotic Intelligence increased by about 0.95 for Geographers; and Manipulation increased by about 0.91 for Airline Pilots. These are the kinds of cases where disagreement is likely to reflect different understandings of what counts as core occupational activity rather than simple scoring error.

The final stage was much more stable overall, but it did not eliminate all movement. Most second-round changes were small, and the average change across all cells was effectively zero. Still, a few cells shifted noticeably. The largest was a late increase in Manipulation for Video Game Designers of about 1.47 levels, followed by increases in Robotic Intelligence and Creativity for Radiologists of about 0.53 and 0.51 respectively. Apart from such isolated cases, the second revision round leaves the overall structure of the ratings largely unchanged.

The rater-level table reinforces the same interpretation (Table A3.3). In the first revision, the observed raters changed on average about 11.2% of their ratings, with individual totals ranging from 2.8% to 25.0%. In the second revision, the average dropped to about 2.5%, with some raters making no changes at all and the largest remaining adjustment reaching 9.4%. The first revision therefore involved genuine reconsideration of the ratings, whereas the second mainly resolved a narrower set of outstanding

disagreements. The direction of change also became less systematic over time. In the first revision, some raters made many more upward than downward adjustments, suggesting a broader recalibration of how demanding certain occupations were judged to be. In the second revision, upward and downward movements were generally more balanced and much smaller in number.

Table A3.3. Direction of evolution of Individual ratings, by rater

A. First Revision

Rater	Total variation	Positive variation	Negative variation
1	9%	8%	1%
2	6%	3%	4%
3	25%	17%	8%
4	12%	7%	6%
5	19%	13%	6%
6	18%	6%	13%
7	4%	3%	1%
8	3%	1%	1%
9	18%	14%	4%
10	3%	2%	1%
11	6%	5%	1%

B. Second Revision

Rater	Total variation	Positive variation	Negative variation
1	2.2%	1.1%	1.1%
2	9.4%	5.0%	4.4%
3	0.0%	0.0%	0.0%
4	1.9%	0.6%	1.4%
5	2.0%	1.9%	0.0%
6	5.6%	0.3%	5.3%
7	1.7%	0.0%	1.7%
8	1.7%	1.1%	0.6%
9	1.1%	0.8%	0.3%
10	1.4%	1.4%	0.0%
11	13.9%	8.7%	5.0%
12	0.0%	0.0%	0.0%

Taken together, the tables suggest that the consensus process worked as intended. The first cycle of discussion and revision played the decisive role in aligning raters' practical interpretation of the capability framework, especially for indicators where the scale descriptors were hard to apply consistently. The second discussion and final revision then served mainly to stabilise the remaining contested cases. The final ratings can therefore be interpreted as a collective expert judgement that is substantially more coherent and internally consistent than the initial individual ratings, while still preserving a limited degree of residual uncertainty for the occupations and indicators that were intrinsically hardest to score.

Annex 4. LLM benchmarking protocol and detailed results

The main goal of creating a Standard of Truth based on human mapping was to build a sound reference to evaluate the quality of LLM's rating process. In this way, it is possible to assess whether LLMs are suitable for mapping human requirements for occupations, thus directly comparing it to an established benchmark. We originally also considered using O*NET abilities and skills to project to the remaining occupations – and we would have done that if LLMs weren't available.

Box A4.1. Selecting, prompting and benchmarking large language models

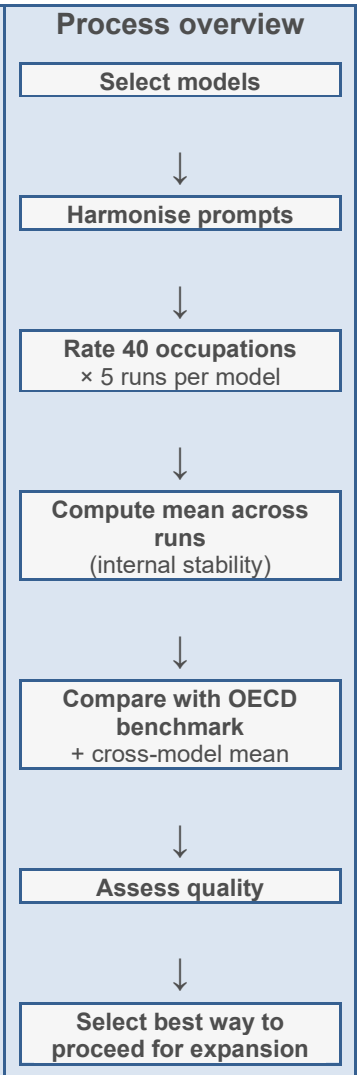
Description of the process

For running the analysis, one large language model from each of four major model families—OpenAI, Anthropic, Google, and Mistral—was selected. ChatGPT 5.2 was used as the reference model, and models from the other families were chosen to ensure the closest possible comparability in terms of scale and processing cost. The models used for the final results were ChatGPT 5.2, Claude 4.6, Gemini 3.1 Pro, and Mistral Large 3. Each model was prompted using mostly the same instructions given to human raters (with small edits to make them suitable for LLM) thereby ensuring consistency across human and model-based evaluations. The prompting protocol adopted in this study was informed by pilot work previously conducted by the OECD using GPT-4.1. All models were accessed through their APIs using their respective python packages.

Each model relied exclusively on the instructions contained in the prompt and on the occupational information supplied in the evaluation materials, including occupational descriptions and task statements derived from O*NET data. This prompt excluded the list of sorted abilities and skills for each occupation, information which was available to human raters, because this additional material did not lead to better rating performance but did significantly increase costs and inference time. In this respect, the model-based exercise was designed to mirror the human rating approach as closely as possible.

The four phases of the benchmarking exercise:

Phase 1. Model selection and prompt harmonisation. One model was selected from each of the four model families. ChatGPT 5.2 served as the reference model, and the remaining models were chosen to maximise comparability in size and processing cost. A common prompting protocol was then developed by adapting the human-rating instructions for model use while preserving their substantive content.



Phase 2. Repeated occupational rating. Each model rated the same set of 40 occupations. To assess reliability, the exercise was repeated over five separate runs for each model.

Phase 3. Stability and benchmarking assessment. Mean scores were computed across the five runs for each model in order to assess internal stability. These model-level averages were then compared with the OECD benchmark, defined as the average human rating. In addition, a cross-model mean was calculated by combining the results across models, generating a further benchmark for comparison with the Standard of Truth.

Phase 4. Quality evaluation and model choice for expansion. The quality of the outputs was evaluated on the basis of their alignment with human judgment and their internal consistency across runs. On this basis, the mean of the models was selected for use in the subsequent expansion process.

Table A4.1 reports internal stability across five runs for each LLM. Mean stability is defined as the average inter-run standard deviation, calculated at the occupation level across the five runs and nine scales; lower values indicate greater model stability. The reported standard deviation captures the extent of heterogeneity in inter-run instability across occupations, showing whether some occupations are more sensitive than others to run-level variation. The minimum and maximum values indicate, respectively, the lowest and highest observed levels of inter-run standard deviation. Following the interpretation proposed in previous studies (Wang & Wang, 2025; Lee et al., 2025; Zeng et al., 2025), mean instability values of 0.10 or below indicate high stability across runs; values between 0.10 and 0.25 suggest that the model is generally stable; values between 0.25 and 0.40 point to a moderate degree of variability; and values above 0.40 indicate high internal variability.

Table A4.1. Internal stability of LLM ratings

Model	Obs.	Mean stability	Std. Dev.	Min	Max
Claude	200	0.09	0.10	0.00	0.37
Gemini	200	0.26	0.13	0.05	0.62
GPT-5.2	200	0.13	0.10	0.00	0.38
Mistral	200	0.08	0.09	0.00	0.40

Accordingly, the results indicate a high degree of internal consistency across all models. The relatively low standard deviation values further suggest that inter-run variability remains limited, supporting the overall robustness of the approach. Although Gemini performs somewhat less strongly than the other models, its mean stability still falls within the range that can be interpreted as moderately stable. Taken together, these findings support the use of all models both for comparison with human average ratings and for the construction of a reliable inter-model average, which can in turn be compared with the OECD mean ratings (see Table A4.1).

In addition, all models are highly correlated on average with the OECD mean (Table A4.2). Correlations are particularly strong for the physical capability scales, such as vision, manipulation and robotic intelligence, which may contain finer and clearer task examples and may therefore be easier for LLMs to rate consistently. By contrast, correlations are somewhat lower for the reasoning capability domains but

creativity, suggesting that the key contrasts in these indicator scales may be more difficult to assess with the occupational information provided. At the aggregate level, however, all models display similar average correlations with the OECD benchmark.

Table A4.2. Pearson correlations between LLM ratings and OECD mean ratings

	Language	Social Interaction	Problem Solving	Creativity	Meta-cognition	Knowledge, Learning and Memory	Vision	Manipulation	Robotic Intelligence	Average
LLMs' Mean	0.86	0.87	0.87	0.92	0.87	0.87	0.96	0.94	0.96	**0.90**
Claude	0.83	0.92	0.85	0.89	0.82	0.86	0.93	0.93	0.86	**0.88**
Mistral	0.85	0.84	0.73	0.92	0.85	0.81	0.87	0.87	0.87	**0.85**
Gemini	0.84	0.76	0.83	0.88	0.71	0.81	0.93	0.88	0.95	**0.84**
GPT	0.80	0.78	0.82	0.83	0.79	0.83	0.91	0.90	0.94	**0.84**
Average	**0.83**	**0.83**	**0.82**	**0.89**	**0.81**	**0.84**	**0.92**	**0.91**	**0.91**	**0.86**

As an additional measure of proximity, and to compare LLM performance with individual human ratings, Table A4.3 reports the root mean squared error (RMSE) of each individual rating relative to the standard of truth, treating each LLM as if it were a single human rater. RMSE is preferred to MSE because it expresses prediction error in the same units as the original variable, thereby facilitating substantive interpretation. The results indicate that the RMSE values obtained for each LLM are sufficiently close to the OECD mean and broadly comparable to those observed among human raters. This degree of proximity suggests that LLM-generated ratings are as credible as the final ratings produced by human assessors through the three-stage Delphi process, thereby providing further support for the consistency and reliability of LLM-based evaluations.

Table A4.3. Roots MSE as compared to OECD mean

Indicator	Rater 1	**LLMs average**	Rater 2	**GPT**	**Claude**	Rater 3	**Mistral**	Rater 4	Rater 5	**Gemini**	Rater 6	Rater 7	Rater 8	Rater 9	Rater 10	Rater 11	**Average**
Language	0.44	0.51	0.57	0.65	0.62	0.72	0.61	0.75	0.59	0.61	0.71	0.75	0.80	0.77	0.98	1.02	**0.68**
Social Interaction	0.69	0.61	0.59	0.72	0.50	0.88	0.71	0.65	0.71	0.86	0.57	0.92	0.92	0.69	1.02	1.13	**0.74**
Problem Solving	0.52	0.53	0.57	0.59	0.54	0.68	0.66	0.61	0.64	0.71	0.85	0.72	1.06	0.77	0.65	0.99	**0.68**
Creativity	0.51	0.50	0.62	0.55	0.59	0.78	0.59	0.72	0.73	0.69	0.81	0.98	0.55	0.83	0.71	0.63	**0.65**
Metacognition	0.40	0.61	0.53	0.68	0.64	0.59	0.62	0.62	0.58	0.74	0.69	0.94	0.95	0.75	0.54	0.77	**0.66**
Knowledge, Learning and Memory	0.47	0.58	0.56	0.66	0.64	0.67	0.55	0.65	1.38	0.93	0.80	0.68	0.61	0.87	0.69	1.13	**0.71**
Vision	0.43	0.48	0.56	0.73	0.71	0.54	0.73	0.77	0.72	0.83	0.88	0.66	0.88	0.78	0.94	1.18	**0.75**
Manipulation	0.47	0.60	0.91	0.76	0.91	0.73	0.87	0.68	0.65	0.77	0.88	0.79	0.67	1.09	0.92	1.33	**0.81**
Robotic Intelligence	0.68	0.55	0.64	0.56	1.03	0.64	0.94	0.86	0.70	0.84	0.81	0.88	0.95	0.92	1.01	0.94	**0.80**
Average	**0.51**	**0.55**	**0.62**	**0.66**	**0.69**	**0.69**	**0.70**	**0.70**	**0.75**	**0.77**	**0.78**	**0.81**	**0.82**	**0.83**	**0.83**	**1.01**	**0.72**

Taken together, these three proximity measures provide consistent evidence supporting the robustness of the LLM-based approach. Across these complementary tests, the models display a satisfactory degree of internal consistency and generate ratings that remain closely aligned with the OECD benchmark. Particularly noteworthy is the fact that the mean of all LLM ratings emerges as the estimate closest to the OECD mean, outperforming the individual models in terms of overall proximity. This suggests that aggregating model outputs helps reduce model-specific variation and yields a more robust approximation of the reference ratings. On this basis, using the inter-model average appears to be the most appropriate strategy for extending the analysis beyond the initial set of 40 occupations to the full occupational space, offering a method that is both analytically reliable and closely aligned with human ratings.

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